

Computational Structures and the Quantum Information Architecture of Spacetime

Research Statement

An empirical investigation of numerical methods that spontaneously generate mathematical structures matching fundamental cosmological parameters, revealing connections between computational complexity, information theory, and physical law.

Contents

1	Introduction	1
2	The Physical System and Computational Challenge	1
3	Methodological Framework	2
3.1	The three-argument structure	2
3.2	From elliptic integrals to transfer functions	3
4	The Scale Alignment and Its Implications	3
4.1	Connection to general relativity	3
4.2	Operational bridge construction	4
5	Empirical Diagnostics and Verification	4
5.1	Continuous diagnostic: The ξ statistic	4
5.2	Discrete diagnostic: The Squarity index	5
5.3	The role of computational time	5
6	Convergence to Perfect-Fluid Limits	5
7	Connections to Quantum Information Frameworks	6
7.1	Information-preserving embeddings	6
7.2	Computational resources as physical coordinates	6
7.3	Topological invariants and discrete diagnostics	6
8	Falsifiable Predictions and Experimental Tests	7
9	Research Trajectory	7
10	Conclusion	8

1 Introduction

The quantum information structure of spacetime represents one of the most promising frameworks for unifying gravitational physics, quantum field theory, and information theory. This perspective posits that the geometric and topological properties we observe in spacetime may not be fundamental but rather emergent from deeper computational and information-theoretic substrates. From holographic principles in string theory to tensor network approaches in condensed matter physics, evidence accumulates that spacetime geometry encodes—or is encoded by—patterns of quantum entanglement and information flow.

Empirical Observation

A data-centric framework developed for biomass gasification systems employs elliptic integrals to achieve mass balance closure. The gains returned by this method consistently fall in the 10^{-47} to 10^{-60} range—the numerical territory occupied by the cosmological constant in Einstein’s field equations.

This observation raises fundamental questions: When numerical methods applied to complex physical systems spontaneously generate mathematical objects whose characteristic scales match fundamental cosmological parameters, what does this reveal about the relationship between computational structures and physical theories? How do information-preserving embeddings, algorithmic complexity measures, and topological invariants—all central to quantum information approaches—manifest in classical non-equilibrium systems? Can data-driven methods illuminate connections between conservation laws and geometric structures that traditional approaches obscure?

This statement presents the methodological framework, empirical evidence, structural diagnostics, and falsifiable predictions that make this line of investigation tractable. The aim is neither to claim immediate resolution of fundamental questions nor to propose speculative connections, but rather to establish a rigorous empirical basis for systematic inquiry into how computational methods for multi-physics systems may encode information about deeper organizational principles in nature.

2 The Physical System and Computational Challenge

Fluidized bed gasification of biomass converts solid carbonaceous material into synthetic gas through thermochemical reactions in a turbulent multiphase environment. The system presents considerable modeling difficulty: fuel heterogeneity (varying moisture, ash, and volatile content), non-equilibrium kinetics (competing reaction pathways with temperature-dependent rates), and coupled transport phenomena (simultaneous mass, momentum, and energy exchange across gas-solid interfaces) create a problem that resists first-principles characterization.

Traditional computational approaches solve coupled systems of partial differential equations describing conservation of mass, momentum, energy, and chemical species. These require closure models for turbulent mixing, uncertain boundary conditions at moving particle surfaces, and empirical correlations for heat and mass transfer coefficients. The resulting computations are intensive, and

model uncertainties compound through the coupling structure.

The framework under investigation takes an alternative path: rather than solving conservation equations directly, it constructs a representation that achieves the same outcome—mass closure in the fluid field—from experimental data through a sequence of transformations culminating in an elliptic integral evaluation. This approach is constructive in a specific sense: it builds the same closure object (a transfer function with physically admissible poles, zeros, and gain) that conservation-based methods target, but does so from observed system behavior rather than postulated governing equations.

3 Methodological Framework

3.1 The three-argument structure

The elliptic integral framework employs three quantities as arguments:

- **Temperature (T):** Absolute temperature serves as the primary thermodynamic state variable. Its inclusion requires no special justification—it is the most fundamental descriptor of thermal state and enters naturally into any energy-based formulation.
- **Specific computational time (sct):** This metric emerges from matrix shearing operations applied to the computational representation of the system. It quantifies algorithmic stiffness—the resistance of the numerical problem to iterative solution. In reactive flow systems, stiffness correlates strongly with thermodynamic non-equilibrium: conditions far from equilibrium generate numerically stiff problems due to disparate timescales in the coupled rate equations. The sct operationalizes this connection, providing a data-derived proxy for the degree of disequilibrium.

The use of computational time as a physical diagnostic is uncommon. Most applications treat algorithmic cost as a practical constraint divorced from system physics. Here, it functions as an independent coordinate capturing aspects of the process state that temperature alone cannot describe. This suggests a deeper relationship between information-theoretic resources (algorithmic complexity) and physical state spaces—a connection central to quantum information approaches to fundamental physics.

- **Composite parameter (κ):** This quantity consolidates diverse factors influencing the process: biomass composition (cellulose, hemicellulose, lignin fractions), particle characteristics (size distribution, density), and operating conditions (fluidization velocity, bed geometry, steam-to-fuel ratio). These variables vary across orders of magnitude and lack natural commensurability.

The composite parameter construction employs canonical correlation analysis to identify stable patterns of co-variation, geometric projections to establish dimensional compatibility, and neural-statistical synthesis to preserve case identity while achieving scale equitability. The result is an information-conserving embedding: a map from high-variance heterogeneous inputs to a common representational space where integration becomes meaningful.

This construction respects a fundamental principle: an integral represents a sum, and meaningful summation requires entities that inhabit compatible spaces. The embedding ensures this compatibility not through arbitrary normalization but through structure-preserving transformation that maintains distinguishability while establishing commensurability.

3.2 From elliptic integrals to transfer functions

The complete elliptic integral of the first kind, implemented via the arithmetic-geometric mean (AGM) algorithm, accepts the triplet (T, sct, κ) and returns a linear time-invariant system:

$$H(s) = k \frac{\prod_{i=1}^{n_z} (s - z_i)}{\prod_{j=1}^{n_p} (s - p_j)}, \quad (1)$$

where z_i are zeros (numerator roots), p_j are poles (denominator roots), and $k > 0$ is the gain.

The use of three-argument elliptic integrals has precedent in celestial mechanics, where they solve orbital dynamics problems. Their application to energy systems represents a transfer of mathematically mature tools into a new physical domain—not model whimsy but a deliberate exploitation of known convergence and stability properties.

Table 1: Transfer function realizations from four representative cases

	Case 1	Case 2	Case 3	Case 4	
Zeros	14×1 complex	16×1 complex	16×1 complex	18×1 complex	<i>Spanning</i>
Poles	15×1 complex	16×1 complex	17×1 complex	18×1 complex	
Gain k	1.45×10^{-47}	3.98×10^{-53}	1.92×10^{-58}	1.26×10^{-60}	

hardwood, agricultural residue, and energy crop feedstocks at varying temperatures (700-850°C) and steam ratios (0.3-0.7). System orders vary modestly while gains span thirteen orders of magnitude.

The zeros and poles emerge as complex conjugate pairs satisfying standard stability criteria for causal systems. The gains vary over thirteen orders of magnitude yet consistently occupy the 10^{-47} to 10^{-60} band. Both features—the constrained pole-zero structure and the systematic gain magnitude—prove consequential for establishing empirical correlations with process observables.

4 The Scale Alignment and Its Implications

4.1 Connection to general relativity

In general relativity, fluid solutions are exact solutions to Einstein’s field equation:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}, \quad (2)$$

where Λ is the cosmological constant representing vacuum energy density. The term $\Lambda g_{\mu\nu}$ contributes a spatially uniform gravitational effect throughout spacetime.

The gains computed by the elliptic integral framework occupy the same numerical band as Λ

values under appropriate unit normalization. This is not claimed as a universal identity independent of dimensional analysis, but rather as an operational correspondence: there exists a unit-consistent normalization under which k maps to the role Λ plays in the field equation.

What prevents this from being dismissed as numerology is the dual functionality of k . The same constant that achieves mass balance closure in the thermochemical system also anchors correlations with experimental performance metrics. It serves simultaneously as a computational output necessary for conservation and as an empirical predictor of system behavior. This dual role suggests the quantity captures something structural rather than serving as a free-floating fit parameter.

4.2 Operational bridge construction

Define $\Lambda_S \equiv k$ for each sample and Λ_G as the geometric mean across the ensemble. Construct the dimensionless log-deviation:

$$\xi \equiv \ln \left(\frac{\Lambda_S}{\Lambda_G} \right). \quad (3)$$

This statistic is two-tailed: $\xi > 0$ indicates above-mean behavior, $\xi < 0$ below-mean. Local variations then take the form

$$\Lambda(x) = \Lambda_G e^{\xi(x)}, \quad (4)$$

providing a map from experimental fluid behavior to a general-relativistic source term structure. The formulation captures spatial, temporal, or operating-point variations in the constitutive constant without requiring the absolute scale to match cosmological measurements—only that the relative structure and the functional form correspond.

5 Empirical Diagnostics and Verification

Two quantities extracted from the transfer function realizations exhibit systematic relationships with process performance:

5.1 Continuous diagnostic: The ξ statistic

Product gas yield demonstrates monotonic dependence on $\xi = \ln(\Lambda_S/\Lambda_G)$ across operating regimes. This relationship emerges at the bin level—aggregating over sufficiently many cases to average measurement noise while preserving trends. Higher yields correspond to systematic deviations in one direction from the geometric mean; lower yields to deviations in the other.

The correlation is not fitted. The ξ statistic derives from the gain magnitude, which itself emerges from the elliptic integral evaluation constrained to achieve mass closure. That this same quantity independently tracks experimental yield provides empirical anchor for the framework's physical validity.

5.2 Discrete diagnostic: The Squarity index

Define the Squarity index as $S = \mathbb{K}[n_z \neq n_p]$, indicating whether pole and zero counts differ. This binary indicator correlates with carbon conversion efficiency and exhibits "locking" behavior at high conversion levels: S stabilizes and ceases to flip between values under small perturbations.

The locking suggests a topological transition in the space of admissible transfer function realizations. At high conversion, the system enters a regime where the structural invariant (equal or unequal pole-zero counts) becomes insensitive to operational variations—analogue to topological phases in condensed matter systems where discrete invariants remain stable under continuous deformations.

5.3 The role of computational time

As operating conditions smooth—better mixing, longer residence times, narrower temperature gradients—the set metric decreases and the distribution of ξ collapses toward zero. Mass closure accuracy improves superlinearly in this regime, consistent with known acceleration of AGM convergence near low eccentricity in elliptic integral evaluation.

This behavior suggests eccentricity in the mathematical formulation serves as a geometric measure of thermodynamic non-equilibrium. Systems far from ideal behavior (large temperature gradients, incomplete reactions, phase segregation) correspond to high eccentricity; systems approaching equilibrium correspond to low eccentricity with attendant convergence acceleration.

6 Convergence to Perfect-Fluid Limits

Define a composite non-equilibrium index:

$$\epsilon \equiv \text{median}(|\xi|) + \alpha \text{Var}(\text{sct}) + \beta \mathcal{K}[S = 1], \quad (5)$$

with weights $\alpha, \beta > 0$. As $\epsilon \rightarrow 0$ corresponding to equilibrium conditions, smooth fields, and saturated pole-zero interlacing—the following properties converge:

- Mass closure residuals vanish
- The ξ distribution collapses (median and variance approach zero)
- The Squarity index locks at $S = 0$ (equal pole and zero counts)
- The set metric contracts toward minimal values

In this limit, the approximate solution from the data-centric framework coincides with exact solutions for perfect fluids—Euler fluids in classical mechanics or the stress-energy tensor for perfect fluids in general relativity with constant Λ . The convergence is structural: the transfer function degenerates to a form consistent with idealized physics where dissipation vanishes and local equilibrium holds.

This establishes a theoretical pathway connecting data-derived representations to limiting cases grounded in fundamental theory. It provides operational meaning to "approaching equilibrium"—not

a vague qualitative statement but a measurable trajectory in (ξ, sct, S) space with quantifiable convergence rates.

7 Connections to Quantum Information Frameworks

Three aspects of this work connect directly to questions central in quantum information approaches to spacetime:

7.1 Information-preserving embeddings

The composite parameter construction employs canonical correlation and geometric regularization to create low-distortion maps from high-dimensional heterogeneous data to lower-dimensional manifolds. This parallels proposals where spacetime geometry emerges from entanglement patterns in quantum states. Both scenarios require transformations that compress dimension while maintaining distinguishability and preserving structural relationships.

In quantum gravity approaches, the question is how bulk spacetime geometry emerges from boundary quantum information. Here, the question is how process-level complexity (dozens of coupled variables with disparate scales) maps into a single parameter that meaningfully enters an integral formulation. Success in the thermochemical context demonstrates that information-conserving dimensional reduction can work in classical non-equilibrium settings, suggesting the technique may be more broadly applicable than typically assumed.

7.2 Computational resources as physical coordinates

The sct metric establishes a direct link between algorithmic complexity (iterative solution cost) and physical state (thermodynamic non-equilibrium). Such connections remain uncommon. The result here—that computational stiffness serves as a legitimate thermodynamic coordinate entering fundamental equations alongside temperature—strengthens the case that information-theoretic resources and physical state spaces share deep connections.

In quantum information approaches to gravity, computational complexity of state preparation or dynamics simulation plays central roles. Circuit complexity measures distance in the space of quantum states; algorithmic cooling protocols manipulate entropy; quantum error correction relates to bulk-boundary correspondence. The patterns observed here suggest analogous structures may operate in classical non-equilibrium thermodynamics, with algorithmic stiffness serving as a diagnostic for departure from idealized limiting behavior.

7.3 Topological invariants and discrete diagnostics

The Squarity index exhibits discrete transitions and locking behavior reminiscent of topological phases. Topological invariants provide robust diagnostics precisely because they count global features rather than measure local properties—they remain stable under continuous deformations. Their appearance in thermochemical conversion dynamics via purely data-driven methods suggests

topological characterization may be more general than currently recognized.

In proposed quantum gravity frameworks, topology changes in spacetime manifolds play fundamental roles (baby universe formation, black hole evaporation, wormhole physics). Observing topology-like transitions in engineered systems through data-driven methods opens possibilities for studying such phenomena in experimentally accessible contexts where controlled perturbations and systematic measurements become feasible.

8 Falsifiable Predictions and Experimental Tests

The framework generates specific expectations amenable to controlled testing:

- **Cross-process portability:** Applying the identical pipeline to pyrolysis, torrefaction, or combustion should preserve three features: (i) mass closure via $H(s)$ of the same structural form, (ii) monotonic ξ -yield correlation, and (iii) S functioning as a discrete efficiency marker. Only the geometric mean Λ_G should require recalibration. Failure would indicate gasification-specific results rather than general principles.
- **Ablation tests:** Removing sct from the integral arguments should degrade mass closure and break ξ -yield monotonicity. Replacing canonical correlation with random embedding should destroy case identity preservation. Eliminating geometric regularization should cause the composite parameter to lose information-conserving properties. Each ablation targets a specific structural claim with predicted failure modes.
- **Convergence quantification:** Constructing operating sequences that smoothly approach equilibrium should produce measurable trajectories in (ξ, sct, S) space. Mass closure error should decrease superlinearly as conditions smooth (AGM acceleration). The Squarity index should lock at high conversion. The sct metric should contract. Quantifying these rates tests the convergence framework.
- **Field reconstruction:** Mapping $\Lambda(x) = \Lambda_G e^{\xi(x)}$ over reactor spatial coordinates should produce smooth fields with physically plausible gradients. Spurious oscillations or unphysical discontinuities would indicate ξ captures statistical noise rather than genuine spatial structure.
- **Noise robustness:** Moderate observational noise should perturb individual zeros and poles while leaving ξ (a ratio to geometric mean) statistically stable. Yield predictions should degrade gracefully rather than collapsing catastrophically, demonstrating the logarithmic formulation provides natural regularization.

9 Research Trajectory

The immediate research trajectory follows directly from the falsification framework outlined above. Testing cross-process portability requires assembling pyrolysis and torrefaction datasets with comparable measurement quality and applying the pipeline without modification. Quantifying

ablation effects demands systematic component removal with careful measurement of degradation in mass closure, correlation strength, and diagnostic stability. Measuring convergence rates requires designing operating sequences that traverse from high to low non-equilibrium while maintaining measurement consistency.

Beyond immediate validation, three deeper questions structure longer-term investigation:

1. **Eccentricity as disequilibrium measure:** Formalizing the connection between elliptic integral eccentricity and thermodynamic non-equilibrium distance. Does eccentricity provide a universal metric for departure from perfect-fluid behavior? Can it be related to standard measures of irreversibility (entropy production, exergy destruction, Carnot efficiency)?
2. **Information geometry:** Does the composite parameter construction define a natural Riemannian metric on the space of operating conditions? If so, do geodesics in this space correspond to optimal process trajectories (minimal dissipation, maximum conversion, fastest equilibration)? Can variational principles be formulated where physical processes extremize action functionals defined on the embedded manifold?
3. **Universal computational structures:** What minimal requirements must a multi-physics problem satisfy for elliptic integral closure to succeed? Is it conservation law structure, scale separation, or something more subtle about how information flows through coupled field equations? Characterizing these requirements would indicate whether the approach addresses a broad class of problems or remains specific to reactive flows.

10 Conclusion

This research statement presents an empirical observation that merits systematic investigation: a data-centric framework for complex thermochemical processes spontaneously generates transfer function gains matching cosmological constant scales. This alignment is accompanied by structural diagnostics that independently correlate with experimental observables and convergence properties that connect to perfect-fluid limits in fundamental theory.

The observation raises questions about relationships between computational structures and physical theories. When numerical methods generate mathematical objects whose scales match fundamental parameters, when algorithmic complexity proxies physical state, when topological invariants diagnose phase-like transitions—these patterns suggest connections between information-theoretic frameworks and physical law that merit rigorous investigation.

The quantum information structure of spacetime provides a natural context for this investigation. Proposals in that domain emphasize emergence of geometry from information, computational complexity as physical observable, and topological invariants as robust diagnostics. The patterns observed here in classical non-equilibrium thermodynamics exhibit striking parallels, suggesting these may be general features of how mathematical structure relates to physical description rather than artifacts specific to quantum gravity.

Whether these connections ultimately illuminate fundamental questions about spacetime architecture or remain useful but domain-specific techniques will be determined through systematic empirical investigation. The framework is falsifiable, the predictions are specific, and the experimental tests are tractable. The next steps are clear: test portability, quantify ablations, measure convergence, reconstruct fields. The observations are documented, the implications are constrained by appropriate epistemic humility, and the path forward follows directly from established scientific method.

Elliptic Analog Low-Pass Prototype: A Complete Zero–Pole–Gain Calculation

$$N = 293, \quad R_p = 8.92 \times 10^{-5} \text{ dB}, \quad R_s = 973.15 \text{ dB}$$

Contents

1	Input Provenance and Problem Statement	2
1.1	The case-1 trio	2
1.2	Signature-valid interpretation	2
1.3	Objective and output cardinalities	2
2	Notation and Symbol Table	4
3	Ripple and Stopband Parameters	6
4	The Complete Integral of the First Kind	7
5	The Complementary Integral and the AGM	8
6	The Nomes	9
7	Prototype Parameter Recovery	10
8	The Jacobi Amplitude	11
9	The Jacobi Elliptic Functions	12
10	Zero Construction	13
11	Pole Construction	14
12	Gain	15
13	Algorithm Summary	16
14	Dependency Chains	17

15 Validation	18
16 Glossary	19
A AGM Iteration Table for $K'(m_1)$	21
B Representative Grid and Output Entries	22

Chapter 1

Input Provenance and Problem Statement

1.1 The case-1 trio

The three governing quantities originate from a single source record. Of its three operations,

$$\text{Operation 1} = 1.035598, \quad \text{Operation 2} = 1.229445, \quad \text{Operation 3} = 8.92 \times 10^{-5},$$

the third is the timed quantity, so the sample computation time is

$$\text{sct} = 8.92 \times 10^{-5} \text{ s.}$$

The order is the sum of five consecutive Fibonacci-like values,

$$A = 18 + 29 + 47 + 76 + 123 = 293,$$

and the temperature, $T = 700^\circ\text{C}$, expressed in kelvin, is

$$T_K = 700 + 273.15 = 973.15 \text{ K.}$$

The resulting trio is $[\text{sct}, A, T_K] = [8.92 \times 10^{-5}, 293, 973.15]$.

1.2 Signature-valid interpretation

The function realized is the elliptic analog prototype $[z, p, k] = \text{ellipap}(N, R_p, R_s)$, whose first argument must be an integer order. The trio is therefore read as

$$N = 293, \quad R_p = 8.92 \times 10^{-5} \text{ dB}, \quad R_s = 973.15 \text{ dB.}$$

1.3 Objective and output cardinalities

We construct the zero-pole-gain representation

$$H(s) = g \frac{\prod_i (s - z_i)}{\prod_i (s - p_i)}. \quad (1.1)$$

The pole count equals the order; because N is odd, one zero lies at infinity, so the finite-zero count is one less:

$$N_p = N = 293, \quad N_z = N - 1 = 292. \quad (1.2)$$

Role. These counts are the targets every later step must reproduce.

Chapter 2

Notation and Symbol Table

Throughout, m is the elliptic *parameter* and k_{mod} the *modulus*, with $m = k_{\text{mod}}^2$. The gain is named g , so the letter k is never overloaded. In $F(\phi \mid m)$ the bar separates the amplitude ϕ from the fixed parameter m and is not a conditional operator; the prime in $K'(\cdot)$ denotes the complementary parameter, not differentiation. In MATLAB/SciPy these map to `ellipap`, `ellipk`, `ellipkm1`, `ellipj`, and `atan2`.

Symbol	Meaning	Definition / value
N	filter order	293
R_p	passband ripple (dB)	8.92×10^{-5}
R_s	stopband attenuation (dB)	973.15
ϵ^2	passband ripple factor	$10^{R_p/10} - 1$
ϵ	ripple factor root	$4.532027135556878 \times 10^{-3}$
$m_1 = k_1^2$	stopband parameter	$\epsilon^2 / (10^{R_s/10} - 1)$
$F(\phi \mid m)$	incomplete integral, first kind	$\int_0^\phi d\theta / \sqrt{1 - m \sin^2 \theta}$
$K(m)$	complete integral, first kind	$F(\pi/2 \mid m)$
$K'(m)$	complementary integral	$K(1 - m)$
$\text{AGM}(a, b)$	arithmetic-geometric mean	limit of $a_{n+1} = \frac{a_n + b_n}{2}$, $b_{n+1} = \sqrt{a_n b_n}$
q_1	stopband nome	$\exp[-\pi K'(m_1)/K(m_1)]$
q	prototype nome	$q_1^{1/N}$
θ_2, θ_3	Jacobi theta constants	series in q
k_{mod}	prototype modulus	$(\theta_2(q)/\theta_3(q))^2$
m	prototype parameter	k_{mod}^2
$\text{am}(u \mid m)$	Jacobi amplitude	inverse of $u = F(\phi \mid m)$
$\text{sn}, \text{cn}, \text{dn}$	Jacobi elliptic functions	$\sin \phi, \cos \phi, \sqrt{1 - m \sin^2 \phi}$
J, J_Z	grid index sets	$\{0, 2, \dots, N-1\}, J \setminus \{0\}$
u_j	grid argument	$jK(m)/N$
s_j, c_j, d_j	Jacobi values at u_j	$\text{sn}, \text{cn}, \text{dn}(u_j \mid m)$
r	pole-offset magnitude	root of $\text{sn} / \text{cn} = 1/\epsilon$ at $1 - m_1$
v_0	pole-offset argument	$K(m)r/(N K(m_1))$
s_v, c_v, d_v	Jacobi values at v_0	$\text{sn}, \text{cn}, \text{dn}(v_0 \mid 1 - m)$
Z, P	finite zeros, poles	complex-conjugate-closed sets

Symbol	Meaning	Definition / value
g	gain	$\text{Re}[\prod(-p_i)/\prod(-z_i)]$

Chapter 3

Ripple and Stopband Parameters

Definition. $\epsilon^2 = 10^{R_p/10} - 1$.

Substitution. $\epsilon^2 = 10^{(8.92 \times 10^{-5})/10} - 1 = 1.000020539269957 - 1$.

Value. $\epsilon^2 = 2.053926995742388 \times 10^{-5}$, $\epsilon = 4.532027135556878 \times 10^{-3}$.

Role. ϵ fixes the equiripple passband edge and later enters the pole offset through $1/\epsilon$.

Definition. $m_1 = \frac{\epsilon^2}{10^{R_s/10} - 1}$.

Substitution. $m_1 = \frac{2.053926995742388 \times 10^{-5}}{2.065380155810529 \times 10^{97}}$.

Value. $m_1 = 9.944546963734884 \times 10^{-103}$.

Role. m_1 is the parameter of the first pair of complete integrals. Physically $m_1 = (\text{passband ripple factor})/(\text{stopband ripple})$, its extreme smallness expresses an extremely severe stopband demand relative to the passband ripple.

Chapter 4

The Complete Integral of the First Kind

With the incomplete integral

$$F(\phi | m) = \int_0^\phi \frac{d\theta}{\sqrt{1 - m \sin^2 \theta}}, \quad K(m) = F\left(\frac{\pi}{2} \middle| m\right), \quad K'(m) = K(1 - m). \quad (4.1)$$

For the small parameter m_1 the direct integral is evaluated near $m = 0$ by the series

$$K(m) = \frac{\pi}{2} \left[1 + \frac{1}{4}m + \frac{9}{64}m^2 + \frac{25}{256}m^3 + \cdots \right], \quad (4.2)$$

whose successive corrections at m_1 are

$$\frac{1}{4}m_1 = 2.486136740933721 \times 10^{-103}, \quad \frac{9}{64}m_1^2 = 1.390697076289622 \times 10^{-205}, \quad \frac{25}{256}m_1^3 = 9.604064157979821 \times 10^{-308}$$

all far below displayed precision, giving

$$K(m_1) = 1.570796326794897. \quad (4.3)$$

Role. $K(m_1)$ is the denominator of the nome exponent.

Chapter 5

The Complementary Integral and the AGM

The complementary integral uses the same m_1 but in the opposite regime, since $m_1 + (1 - m_1) = 1$. Here $1 - m_1$ is extremely close to 1, where K has a logarithmic singularity and $K(1) = \infty$; the value $1 - m_1$ must therefore *not* be rounded to 1. It is computed by the arithmetic–geometric mean

$$K(m) = \frac{\pi}{2 \operatorname{AGM}(1, \sqrt{1 - m})} \implies K'(m_1) = K(1 - m_1) = \frac{\pi}{2 \operatorname{AGM}(1, \sqrt{m_1})}, \quad (5.1)$$

where the mean is the common limit of the coupled recurrence

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \sqrt{a_n b_n}, \quad a_0 = 1, \quad b_0 = \sqrt{m_1} = 9.972234936931081 \times 10^{-52}. \quad (5.2)$$

The iteration converges by $n = 10$ (full table in Appendix A):

$$\operatorname{AGM}(1, \sqrt{m_1}) = 0.013219863975783, \quad K'(m_1) = \frac{\pi}{2(0.013219863975783)} = 118.8209144717663.$$

Role. The ratio $K'(m_1)/K(m_1)$ measures the complementary severity of the specification and sets the nome. Note $K(m_1) \neq K(1 - m_1)$: the two integrals share a parameter but live in different regimes.

Chapter 6

The Nomes

Definition. $q_1 = \exp\left[-\pi \frac{K'(m_1)}{K(m_1)}\right]$.

Value. $q_1 = \exp\left[-\pi \frac{118.8209144717663}{1.570796326794897}\right] = 6.215341852334302 \times 10^{-104}$.

Definition. $q = q_1^{1/N}$.

Value. $q = 0.4443848722641027$.

Role. q is the nome of the prototype parameter recovered next.

Chapter 7

Prototype Parameter Recovery

The parameter is recovered from its nome through the Jacobi theta constants

$$\theta_2(q) = 2 \sum_{n \geq 0} q^{(n+1/2)^2} = 1.96808021371183, \quad \theta_3(q) = 1 + 2 \sum_{n \geq 1} q^{n^2} = 1.968121074283701, \quad (7.1)$$

$$k_{\text{mod}} = \left(\frac{\theta_2(q)}{\theta_3(q)} \right)^2 = 0.9999584780141351, \quad m = k_{\text{mod}}^2 = \left(\frac{\theta_2(q)}{\theta_3(q)} \right)^4 = 0.9999169577523455, \quad (7.2)$$

with complete integral

$$K(m) = 6.084480456260613. \quad (7.3)$$

Role. m and $K(m)$ define the Jacobi grid carrying the zeros and poles.

Chapter 8

The Jacobi Amplitude

The amplitude is the inverse of the incomplete integral: from $u = F(\phi | m)$,

$$\phi = \text{am}(u | m) = F^{-1}(u | m), \quad (8.1)$$

which has no elementary closed form except at $m = 0, 1$. Given sn, cn it is recovered quadrant-correctly by $\phi = \text{atan2}(\text{sn}, \text{cn})$, or on the principal branch by $\phi = \sin^{-1} \text{sn}$. Differentiating $u = F(\phi | m)$ gives

$$\frac{d\phi}{du} = \sqrt{1 - m \sin^2 \phi} = \text{dn}(u | m), \quad \text{am}(0 | m) = 0. \quad (8.2)$$

It may be computed directly by Newton iteration on $f(\phi) = F(\phi | m) - u$:

$$\phi_{n+1} = \phi_n - [F(\phi_n | m) - u] \sqrt{1 - m \sin^2 \phi_n}. \quad (8.3)$$

Special cases. At $m = 0$: $F(\phi | 0) = \phi$, so $\text{am}(u | 0) = u$ and $\text{sn} = \sin u$, $\text{cn} = \cos u$, $\text{dn} = 1$. At $m = 1$: $F(\phi | 1) = \ln |\sec \phi + \tan \phi| = \text{artanh}(\sin \phi)$, so $\text{am}(u | 1) = \sin^{-1}(\tanh u)$ and $\text{sn} = \tanh u$, $\text{cn} = \text{dn} = \text{sech } u$. For instance $\text{am}(0.5 | 1) = 0.4803810791337294$ with $\text{sn}(0.5 | 1) = \tanh 0.5 = 0.4621171572600098$.

Role. The amplitude underlies the definitions of $\text{sn}, \text{cn}, \text{dn}$ used everywhere below.

Chapter 9

The Jacobi Elliptic Functions

With $\phi = \text{am}(u \mid m)$,

$$\text{sn}(u \mid m) = \sin \phi, \quad \text{cn}(u \mid m) = \cos \phi, \quad \text{dn}(u \mid m) = \sqrt{1 - m \sin^2 \phi} = \sqrt{1 - m \text{sn}^2(u \mid m)}. \quad (9.1)$$

They obey the identities used for validation,

$$\text{sn}^2 + \text{cn}^2 = 1, \quad \text{dn}^2 + m \text{sn}^2 = 1, \quad (9.2)$$

the derivative relations

$$\frac{d}{du} \text{sn} = \text{cn} \, \text{dn}, \quad \frac{d}{du} \text{cn} = -\text{sn} \, \text{dn}, \quad \frac{d}{du} \text{dn} = -m \text{sn} \, \text{cn}, \quad (9.3)$$

with initial values $\text{sn}(0 \mid m) = 0$, $\text{cn}(0 \mid m) = 1$, $\text{dn}(0 \mid m) = 1$, and the small- u series

$$\text{sn}(u \mid m) = u - \frac{1+m}{6}u^3 + \frac{1+14m+m^2}{120}u^5 + \dots, \quad (9.4)$$

$$\text{cn}(u \mid m) = 1 - \frac{u^2}{2} + \frac{1+4m}{24}u^4 - \frac{1+44m+16m^2}{720}u^6 + \dots, \quad (9.5)$$

$$\text{dn}(u \mid m) = 1 - \frac{m}{2}u^2 + \frac{m(4+m)}{24}u^4 - \frac{m(16+44m+m^2)}{720}u^6 + \dots. \quad (9.6)$$

In MATLAB/SciPy these are obtained as `[sn,cn,dn]=ellipj(u,m)`.

Role. Evaluated on the grid, sn builds the zeros; sn,cn,dn together build the poles.

Chapter 10

Zero Construction

For odd N the grid index set and its zero-bearing subset are

$$J = \{0, 2, 4, \dots, 292\}, \quad J_Z = J \setminus \{0\}, \quad |J_Z| = 146, \quad (10.1)$$

the point $j = 0$ excluded because $\text{sn}(0 \mid m) = 0$ would force an infinite zero. With $u_j = jK(m)/N$ and $s_j = \text{sn}(u_j \mid m)$,

$$z_j = \frac{i}{\sqrt{m} s_j}, \quad j \in J_Z. \quad (10.2)$$

Because $H(s)$ has real coefficients, non-real zeros occur in conjugate pairs; here the finite zeros are purely imaginary, $z_j = +i\omega_j$ with conjugate $z_j^* = -i\omega_j$, so

$$Z = \{z_j : j \in J_Z\} \cup \{z_j^* : j \in J_Z\}, \quad |Z| = 146 + 146 = 292. \quad (10.3)$$

Worked entry ($j = 2$). $u_2 = 0.04153228980382671$, $s_2 = 0.04150842714638451$, $c_2 = 0.9991381538485222$, $d_2 = 0.9991382254490288$, amplitude $\phi_2 = \text{atan2}(s_2, c_2) = 0.04152035588484406$; the series gives $\text{sn}(u_2 \mid m) \approx 0.04150842715787879$ in agreement. Hence $z_2 = i(24.09249380091527)$. In MATLAB: `z=1i./(sqrt(m).*sn).`

Role. Z is the numerator factor set of $H(s)$.

Chapter 11

Pole Construction

The pole offset uses a real magnitude r , the unique positive root of

$$\frac{\operatorname{sn}(r \mid 1 - m_1)}{\operatorname{cn}(r \mid 1 - m_1)} = \frac{1}{\epsilon}, \quad r = 6.08973826362654, \quad (11.1)$$

from which

$$v_0 = \frac{K(m)r}{N K(m_1)} = 0.08050718291099445, \quad (11.2)$$

and the offset Jacobi values at the complementary parameter $1 - m$ are

$$s_v = \operatorname{sn}(v_0 \mid 1 - m) = 0.08042023727098944, \quad c_v = 0.9967610473113793, \quad d_v = 0.999999731465643. \quad (11.3)$$

With $s_j, c_j, d_j = \operatorname{sn}, \operatorname{cn}, \operatorname{dn}(u_j \mid m)$, each pole is

$$p_j = -\frac{c_j d_j s_v c_v + i s_j d_v}{1 - (d_j s_v)^2}, \quad j \in J. \quad (11.4)$$

At $j = 0$, $s_0 = 0$, $c_0 = d_0 = 1$, yielding a single real pole

$$p_0 = -\frac{s_v c_v}{1 - s_v^2} = -0.08068156103000971, \quad (11.5)$$

while the remaining values are complex, $p_j = \alpha_j - i\beta_j$ with $\alpha_j < 0$, appearing with conjugates $p_j^* = \alpha_j + i\beta_j$:

$$P = \{p_0\} \cup \{p_j : j \in J_Z\} \cup \{p_j^* : j \in J_Z\}, \quad |P| = 1 + 146 + 146 = 293. \quad (11.6)$$

Worked entry ($j = 2$). $p_2 = -0.08054165329958819 - 0.04177814710046775i$. In MATLAB: `[sv,cv,dv]=ellipj(v0,1-m)` and `p=-((cn.*dn.*sv.*cv)+1i.*sn.*dv)./(1-(dn.*sv).^2)`.

Role. P is the denominator factor set of $H(s)$.

Chapter 12

Gain

Definition. $g = \operatorname{Re} \left[\frac{\prod_i (-p_i)}{\prod_i (-z_i)} \right]$.

Value. $g = 1.664493218421019 \times 10^{-47}$.

Role. g completes the triple (Z, P, g) defining $H(s)$.

Chapter 13

Algorithm Summary

Input: N, R_p, R_s . **Output:** Z, P, g .

1. $\epsilon^2 = 10^{R_p/10} - 1$.
2. $m_1 = \epsilon^2 / (10^{R_s/10} - 1)$.
3. $K(m_1)$, $K'(m_1) = K(1 - m_1)$ by AGM.
4. $q_1 = \exp(-\pi K'(m_1)/K(m_1))$.
5. $q = q_1^{1/N}$.
6. Solve the degree equation for m via $m = (\theta_2(q)/\theta_3(q))^4$.
7. $K(m)$.
8. $J = \{0, 2, \dots, N-1\}$ for odd N .
9. $u_j = jK(m)/N$.
10. $s_j, c_j, d_j = \text{sn}, \text{cn}, \text{dn}(u_j \mid m)$.
11. Finite zeros $z_j = i/(\sqrt{m} s_j)$ for $j \neq 0$.
12. $r, v_0, s_v, c_v, d_v = \text{sn}, \text{cn}, \text{dn}(v_0 \mid 1 - m)$.
13. Poles p_j .
14. Append conjugates.
15. Gain g .

Chapter 14

Dependency Chains

$$(N, R_p, R_s) \Rightarrow \epsilon^2, \epsilon, m_1 \Rightarrow K(m_1), K'(m_1), q_1 \Rightarrow q, m, K(m) \Rightarrow u_j, s_j, c_j, d_j \Rightarrow Z,$$

and, sharing the early scalars,

$$\epsilon, m_1, K(m_1), K(m), N \Rightarrow r, v_0, s_v, c_v, d_v \Rightarrow P \Rightarrow g.$$

Chapter 15

Validation

The completed triple satisfies every structural requirement:

$$|Z| = 292, \quad |P| = 293, \quad (15.1)$$

$$\operatorname{Re}(p_i) < 0 \quad \forall i \quad (\max_i \operatorname{Re} p_i = -1.38273 \times 10^{-7}), \quad (15.2)$$

$$z_i^* \in Z \text{ whenever } z_i \in Z, \quad p_i^* \in P \text{ whenever } p_i \in P, \quad (15.3)$$

$$s_j^2 + c_j^2 = 1, \quad d_j^2 + m s_j^2 = 1, \quad (15.4)$$

$$g = \operatorname{Re}[\prod_i (-p_i) / \prod_i (-z_i)] = 1.664493218 \times 10^{-47}. \quad (15.5)$$

Negativity of all real parts confirms stability; conjugate closure confirms real coefficients; the Jacobi identities confirm the grid values.

Chapter 16

Glossary

Order N

Number of poles of $H(s)$; here 293 (odd).

Passband ripple R_p

Permitted ripple in dB; 8.92×10^{-5} .

Stopband attenuation R_s

Required attenuation in dB; 973.15.

Ripple factor ϵ

$\sqrt{10^{R_p/10} - 1}$; sets the passband edge and $1/\epsilon$ in the pole offset.

Stopband parameter $m_1 = k_1^2$

$\epsilon^2/(10^{R_s/10} - 1)$; ratio of passband-ripple to stopband-attenuation factors.

Incomplete elliptic integral $F(\phi | m)$

$\int_0^\phi d\theta / \sqrt{1 - m \sin^2 \theta}$; the bar separates amplitude from parameter.

Complete elliptic integral $K(m)$

$F(\pi/2 | m)$; complete first-kind integral.

Complementary integral $K'(m)$

$K(1 - m)$; the prime denotes complementary parameter, not derivative.

Arithmetic–geometric mean $\text{AGM}(a, b)$

Common limit of $a_{n+1} = (a_n + b_n)/2$, $b_{n+1} = \sqrt{a_n b_n}$; used to evaluate K near the $m = 1$ singularity.

Stopband nome q_1

$\exp[-\pi K'(m_1)/K(m_1)]$.

Prototype nome q

$q_1^{1/N}$; nome of the prototype parameter.

Theta constants θ_2, θ_3

$\theta_2 = 2 \sum q^{(n+1/2)^2}$, $\theta_3 = 1 + 2 \sum q^{n^2}$; invert the nome.

Modulus k_{mod}

$(\theta_2/\theta_3)^2$; with $m = k_{\text{mod}}^2$.

Prototype parameter m

$(\theta_2/\theta_3)^4$; the elliptic parameter of the grid.

Jacobi amplitude $\text{am}(u \mid m)$

Inverse of $u = F(\phi \mid m)$; equals $\text{atan2}(\text{sn}, \text{cn})$; satisfies $d \text{ am} / du = \text{dn}$, $\text{am}(0 \mid m) = 0$.

Jacobi functions $\text{sn}, \text{cn}, \text{dn}$

$\sin \phi, \cos \phi, \sqrt{1 - m \sin^2 \phi}$ with $\phi = \text{am}(u \mid m)$; obey $\text{sn}^2 + \text{cn}^2 = 1$ and $\text{dn}^2 + m \text{sn}^2 = 1$.

Grid sets J, J_Z

$J = \{0, 2, \dots, N-1\}$ (odd N); $J_Z = J \setminus \{0\}$, $|J_Z| = 146$.

Grid argument u_j

$jK(m)/N$.

Grid values s_j, c_j, d_j

$\text{sn}, \text{cn}, \text{dn}(u_j \mid m)$.

Pole-offset magnitude r

Root of $\text{sn}(r \mid 1 - m_1) / \text{cn}(r \mid 1 - m_1) = 1/\epsilon$; 6.08973826362654.

Pole-offset argument v_0

$K(m)r/(N K(m_1))$; 0.08050718291099445.

Offset values s_v, c_v, d_v

$\text{sn}, \text{cn}, \text{dn}(v_0 \mid 1 - m)$.

Zeros Z

$z_j = i/(\sqrt{m} s_j)$ plus conjugates; $|Z| = 292$.

Poles P

$p_j = -(c_j d_j s_v c_v + i s_j d_v) / (1 - (d_j s_v)^2)$, real pole at $j = 0$, plus conjugates; $|P| = 293$.

Gain g

$\text{Re}[\prod(-p_i) / \prod(-z_i)]$; $1.664493218 \times 10^{-47}$.

Transfer function $H(s)$

$g \prod(s - z_i) / \prod(s - p_i)$.

MATLAB/SciPy primitives

`ellipap` (prototype), `ellipk` (complete integral), `ellipkm1` (complementary), `ellipj` (Jacobi functions), `atan2` (amplitude).

Appendix A

AGM Iteration Table for $K'(m_1)$

$a_0 = 1$, $b_0 = \sqrt{m_1} = 9.972234936931081 \times 10^{-52}$.

n	a_n	b_n
0	1.000000000000	$9.972234936931 \times 10^{-52}$
1	0.500000000000	$3.157884566752 \times 10^{-26}$
2	0.250000000000	$1.256559701477 \times 10^{-13}$
3	0.125000000000	$1.772399292960 \times 10^{-7}$
4	0.062500088620	$1.488455278540 \times 10^{-4}$
5	0.031324467074	$3.050058799690 \times 10^{-3}$
6	0.017187262937	$9.774531520450 \times 10^{-3}$
7	0.013480897229	$1.296138277060 \times 10^{-2}$
8	0.013221139900	$1.321858801350 \times 10^{-2}$
9	0.013219864007	$1.321986394500 \times 10^{-2}$
10	0.013219863976	$1.321986397580 \times 10^{-2}$

Limit AGM($1, \sqrt{m_1}$) = 0.013219863975783, hence $K'(m_1) = \pi/[2 \text{ AGM}] = 118.8209144717663$.

Appendix B

Representative Grid and Output Entries

j	u_j	$s_j = \text{sn}(u_j \mid m)$	finite zero z_j
2	0.04153228980382671	0.04150842714638451	$i(24.09249380091527)$
4	0.08306457960765342	0.08287407237791083	$i(12.06700111404881)$
292	6.0637143113587	0.9999999820921569	$i(1.000041541618599)$

j	pole p_j
0	-0.08068156103000971 (real)
2	$-0.08054165329958819 - 0.04177814710046775 i$
4	$-0.08012387216450066 - 0.08340979115563503 i$
292	$-1.382727025629448 \times 10^{-7} - 1.000000250858197 i$

CHAPTER 4

METHODOLOGY

4.1 Canonical correlation analysis of biomass properties

As mentioned in previous chapters, our strategy is to first construct a congruent composite set representing each sample of database, and then through performing a series of functional analyses, detect predictors of composition of product gas, carbon conversion and yield. In the process, we utilize a selective set of machine learning algorithms, differential geometry/trigonometry concepts, neural networks, calculus of variations, and work with various mathematical objects, such as scalars, vectors, matrices, polygons, and pairwise relations constructed from statistical analysis of data.

We begin with development of correlations between different biomass property groups using Canonical Component Analysis (CCA):

Composition (C, H, N, O, S), proximate analysis (Volatile matter, Fixed carbon, Ash), structural component (Cellulose, Hemicellulose, Lignin) and ash composition from 38 types of biomass is collected, as given in Tables 12-14. [78]

Table 12. Ultimate and proximate analysis of 38 types of biomass (Page 1 of 2)

No	Name	VM	FC	A	C	O	H	N	S	No
1	Beech bark	73.69	18.56	7.75	51.4	41.8	6	0.7	0.11	1
2	Birch bark	78.49	19.43	2.07	57	35.7	6.7	0.5	0.1	2
3	Eucalyptus bark	78.07	17.16	4.77	48.7	45.3	5.7	0.3	0.05	3
4	Oak sawdust	86.21	13.45	0.34	50.1	43.9	5.9	0.1	0.01	4
5	Oak wood	78.07	21.39	0.53	50.6	42.9	6.1	0.3	0.1	5
6	Olive wood	79.55	17.24	3.21	49	44.9	5.4	0.7	0.03	6
7	Pine bark	73.66	24.45	1.89	53.8	39.9	5.9	0.3	0.07	7
8	Pine chips	72.40	21.65	5.95	52.8	40.5	6.1	0.5	0.09	8
9	Pine pruning	82.32	15.02	2.66	51.9	41.3	6.3	0.5	0.01	9
10	Pine sawdust	83.12	16.77	0.12	51	42.9	6	0.1	0.01	10
11	Spruce bark	73.47	23.36	3.17	53.6	40	6.2	0.1	0.1	11

Table 12. Ultimate and proximate analysis of 38 types of biomass (Page 2 of 2)

No	Name	VM	FC	A	C	O	H	N	S	No
12	Spruce wood	81.14	18.33	0.54	52.3	41.2	6.1	0.3	0.1	12
13	Wood	84.06	15.73	0.22	49.6	44.1	6.1	0.1	0.06	13
14	Wood residue	77.99	16.58	5.43	51.4	41.9	6.1	0.5	0.08	14
15	Bamboo whole	81.61	17.47	0.92	52	42.5	5.1	0.4	0.04	15
16	Reed canary grass	73.46	17.66	8.88	49.4	42.7	6.3	1.5	0.15	16
17	Sweet sorghum	77.20	18.06	4.73	49.7	43.7	6.1	0.4	0.09	17
18	Switchgrass	80.36	14.53	5.11	49.7	43.4	6.1	0.7	0.11	18
19	Barley straw	76.16	18.53	5.31	49.4	43.6	6.2	0.7	0.13	19
20	Oat straw	80.50	13.62	5.88	48.8	44.6	6	0.5	0.08	20
21	Rape straw	77.44	17.85	4.71	48.5	44.5	6.4	0.5	0.1	21
22	Rice straw	64.29	15.58	20.13	50.1	43	5.7	1	0.16	22
23	Straw	73.40	15.75	10.84	48.8	44.5	5.6	1	0.13	23
24	Wheat straw	74.75	18.13	7.12	49.4	43.6	6.1	0.7	0.17	24
25	Almond shells	74.89	21.77	3.34	50.3	42.5	6.2	1	0.05	25
26	Coconut shells	73.74	23.01	3.24	51.1	43.1	5.6	0.1	0.1	26
27	Cotton husks	78.41	18.15	3.44	50.4	39.8	8.4	1.4	0.01	27
28	Hazelnut shells	77.05	21.44	1.51	51.5	41.6	5.5	1.4	0.04	28
29	Olive husks	79.08	18.67	2.25	50	42.1	6.2	1.6	0.05	29
30	Olive pits	77.00	19.91	3.09	52.8	39.4	6.6	1.1	0.07	30
31	Olive residue	67.34	25.50	7.16	58.4	34.2	5.8	1.4	0.23	31
32	Rice husks	62.75	19.24	18.01	49.3	43.7	6.1	0.8	0.08	32
33	Sunflower husks	76.02	20.90	3.08	50.4	43	5.5	1.1	0.03	33
34	Walnut blows	80.78	16.86	2.35	54.9	36.9	6.7	1.4	0.11	34
35	Walnut hulls	79.65	17.47	2.88	55.1	36.5	6.7	1.6	0.12	35
36	Walnut shells	59.33	37.88	2.79	49.9	42.4	6.2	1.4	0.09	36
37	Wood residue	78.48	18.22	3.30	52.4	41.2	6	0.4	0.04	37
38	Straw residue	75.08	16.72	8.20	51.7	41.5	6.3	0.4	0.13	38

Table 13. Structural component of the 38 types of biomass under study. [7] (Page 1 of 2)

Ref	Name	Cellulose	Hemicellulose	Lignin
1	Beech bark	45.2	32.7	22.1
2	Birch bark	50.2	32.8	17
3	Eucalyptus bark	52.7	15.4	31.9
4	Oak sawdust	58.4	31.4	10.2
5	Oak wood	58.4	31.4	10.2
6	Olive wood	25	24.6	50.4
7	Pine bark	48.1	23.5	28.4
8	Pine chips	48.1	23.5	28.4
9	Pine pruning	48.1	23.5	28.4
10	Pine sawdust	45.9	26.4	27.7
11	Spruce bark	43.6	27.4	29

Table 13. Structural component of the 38 types of biomass under study. [7] (Page 2 of 2)

Ref	Name	Cellulose	Hemicellulose	Lignin
12	Spruce wood	47	25.3	27.7
13	Wood	42	22	36
14	Wood residue	50.6	24.7	24.7
15	Bamboo whole	43.9	26.5	29.6
16	Reed canary grass	49.2	39.1	11.7
17	Sweet sorghum grass	50.6	24.7	24.7
18	Switchgrass	48.7	38.4	12.9
19	Barley straw	48.6	29.7	21.7
20	Oat straw	44.8	33.4	21.8
21	Rape straw	54.8	23.2	22
22	Rice straw	52.3	32.8	14.9
23	Straw	44.5	33.2	22.3
24	Wheat straw	44.5	33.2	22.3
25	Almond shells	50.7	28.9	20.4
26	Coconut shells	40.3	27.8	31.9
27	Cotton husks	82.8	17.1	0.1
28	Hazelnut shells	26.6	30	43.4
29	Olive husks	25	24.6	50.4
30	Olive pits	25	24.6	50.4
31	Olive residue	25	24.6	50.4
32	Rice husks	43.8	31.6	24.6
33	Sunflower husks	56.5	28	15.5
34	Walnut blows	28.1	26.6	45.3
35	Walnut hulls and blows	28.1	26.6	45.3
36	Walnut shells	28.1	26.6	45.3
37	Wood-agricultural residue	46.8	30.4	22.8
38	Wood-straw residue	44.8	33.4	21.8

Table 14. Ash composition of the 38 types of biomass under study [78] (Page 1 of 2)

Ref	Name	SiO2	CaO	K2O	P2O5	Al2O3	MgO	Fe2O3	SO3	Na2O	TiO2
1	Beech bark	12.4	68.2	2.6	2.3	0.12	11.5	1.1	0.8	0.9	0.1
2	Birch bark	4.38	69.06	8.99	4.13	0.55	5.92	2.24	2.75	1.85	0.13
3	Eucalyptus bark	10.04	57.74	9.29	2.35	3.1	10.91	1.12	3.47	1.86	0.12
4	Oak sawdust	29.93	15.56	31.99	1.9	4.27	5.92	4.2	3.84	2	0.39
5	Oak wood	48.95	17.48	9.49	1.8	9.49	1.1	8.49	2.6	0.5	0.1
6	Olive wood	10.24	41.47	25.16	10.75	2.02	3.03	0.88	2.65	3.67	0.13
7	Pine bark	9.2	56.83	7.78	5.02	7.2	6.19	2.79	2.83	1.97	0.19
8	Pine chips	68.18	7.89	4.51	1.56	7.04	2.43	5.45	1.19	1.2	0.55
9	Pine pruning	7.76	44.1	22.32	5.73	2.75	11.33	1.25	4.18	0.42	0.17
10	Pine sawdust	9.71	48.88	14.38	6.08	2.34	13.8	2.1	2.22	0.35	0.14

Table 14. Ash composition of the 38 types of biomass under study [78] (Page 2 of 2)

Ref	Name	SiO ₂	CaO	K ₂ O	P ₂ O ₅	Al ₂ O ₃	MgO	Fe ₂ O ₃	SO ₃	Na ₂ O	TiO ₂
11	Spruce bark	6.13	72.39	7.22	2.69	0.68	4.97	1.9	1.88	2.02	0.12
12	Spruce wood	49.3	17.2	9.6	1.9	9.4	1.1	8.3	2.6	0.5	0.1
13	Wood	23.15	37.35	11.59	2.9	5.75	7.26	3.27	4.95	2.57	1.2
14	Wood residue	53.15	11.66	4.85	1.37	12.64	3.06	6.24	1.99	4.47	0.57
15	Bamboo whole	9.92	4.46	53.38	20.33	0.67	6.57	0.67	3.68	0.31	0.01
16	Reed canary	84.92	3.31	2.93	3.88	1.32	1.42	1.04	1.04	0.09	0.05
17	Sweet sorghum	66.85	10.41	9.49	3.47	0.81	3.12	0.58	3.47	1.74	0.06
18	Switchgrass	66.25	10.21	9.64	3.92	2.22	4.71	1.36	0.83	0.58	0.28
19	Barley straw	50.78	9.89	28.18	2.97	0.67	2.87	0.95	2.22	1.39	0.08
20	Oat straw	37.79	12.03	26.84	6.14	4.69	4.45	2.17	4.93	0.72	0.24
21	Rape straw	40.8	30.68	13.45	2.22	5.45	2	2	2.67	0.44	0.29
22	Rice straw	77.2	2.46	12.59	0.98	0.55	2.71	0.5	1.18	1.79	0.04
23	Straw	57.14	6.7	25.82	2.74	0.76	1.67	0.53	3.89	0.7	0.05
24	Wheat straw	50.35	8.21	24.89	3.54	1.54	2.74	0.88	4.24	3.52	0.09
25	Almond shells	16.96	11.55	53.48	4.93	2.99	4.51	2.78	0.93	1.76	0.11
26	Coconut shells	66.75	2.41	8.48	1.54	8.48	1.54	6.16	0.01	4.62	0.01
27	Cotton husks	10.93	20.95	50.2	4.05	1.32	7.59	1.92	1.72	1.31	0.01
28	Hazelnut shells	33.7	15.4	30.4	3.2	3.1	7.9	3.8	1.1	1.3	0.1
29	Olive husks	32.7	14.5	4.3	2.5	8.4	4.2	6.3	0.6	26.2	0.3
30	Olive pits	21.48	19.97	16.44	9.71	5.95	3.84	4.25	2.3	15.77	0.29
31	Olive residue	22.26	12.93	42.79	6.09	4.1	5.84	1.99	3.73	0.12	0.15
32	Rice husks	94.48	0.97	2.29	0.54	0.21	0.19	0.22	0.92	0.16	0.02
33	Sunflower husks	23.66	15.31	28.53	7.13	8.75	7.33	4.27	4.07	0.8	0.15
34	Walnut blows	6.41	27.64	34.67	10.28	2.25	14.34	1.05	2.33	0.92	0.11
35	Walnut hulls	8.29	20.03	39.65	7.52	2.92	16.21	1.37	2.71	1.19	0.11
36	Walnut shells	23.32	16.72	33.03	6.21	2.4	13.51	1.5	2.2	1	0.1
37	Wood-residue	37.18	25.7	7.76	2.22	11.07	4.77	5.77	2.03	2.57	0.93
38	Straw residue	57.83	11.51	6.67	1.08	9.77	2.66	4.97	1.88	3.11	0.52

The main goal is to incorporate the consideration of the effect of ultimate, proximate, structural and ash content in predicting the composition of product gas. In addition, this method develops relationships between different types of biomass analyses

utilizing statistically inferred features that express various ranges of irregularity in biomass. In the following sections, the step by step of this statistical process is described for Elemental composition, Proximate analysis, and structural component. The ash composition is treated separately due to more heterogeneity, the main part of methodology is presented in chapter 6 and needs to be completed in future.

“CCA is a method for exploring the relationships between two multivariate sets of variables (vectors), all measured on the same individual [79].”

$$X = [X_1 \ X_2 \ ..X_p \], Y = [Y_1 \ Y_2 \ ..Y_q \] \quad (27)$$

$$U_1 = a_{11}X_1 + a_{12}X_2 + \dots + a_{1p}X_p$$

$$U_2 = a_{21}X_1 + a_{22}X_2 + \dots + a_{2p}X_p$$

...

$$U_p = a_{p1}X_1 + a_{p2}X_2 + \dots + a_{pp}X_p$$

&

$$V_1 = b_{11}Y_1 + b_{12}Y_2 + \dots + b_{1q}Y_q$$

$$V_2 = b_{21}Y_1 + b_{22}Y_2 + \dots + b_{2q}Y_q$$

...

$$V_p = b_{p1}Y_1 + b_{p2}Y_2 + \dots + b_{pq}Y_q$$

(U_i, V_i)

(28)

In this setup, (U1, V1) is the first canonical variate pair, and (U2, V2) would be the second canonical variate pair and so on... With $p \leq q$ there are p canonical covariate pair

The objective is to find linear combinations that maximize the correlations between the members of each canonical variate pair.

“We can compute the variance of U_i variables using the following expression [79]”:

$$var(U_i) = \sum_{k=1}^p \sum_{l=1}^p a_{ik} a_{il} cov(X_k, X_l) \quad (29)$$

The coefficients a_{i1} through a_{ip} that appear in the double sum are the same coefficients that appear in the definition of U_i . The covariances between the k th and l th X-variables are multiplied by the corresponding coefficients a_{ik} and a_{il} for the variate U_i .

Similar calculations can be made for the variance of V_i as shown below [79]:

$$var(V_i) = \sum_{k=1}^p \sum_{l=1}^q b_{ik} b_{il} cov(Y_k, Y_l) \quad (30)$$

Then calculate the covariance between U_i and V_i as [79]:

$$Cov(U_i, V_i) = \sum_{k=1}^p \sum_{l=1}^q a_{ik} b_{il} cov(X_k, Y_l) \quad (31)$$

The correlation between U_i and V_i is calculated by taking the covariance between two variables and divide it by the square root of the product of the variances:

$$\frac{cov(U_i, V_j)}{\sqrt{var(U_i)var(V_j)}} \quad (32)$$

“The canonical correlation is a specific type of correlation. The canonical correlation for the i^{th} canonical variate pair is simply the correlation between U_i and V_i

$$[80]”: \rho_i^* = \frac{cov(U_i, V_j)}{\sqrt{var(U_i)var(V_j)}} \quad (33)$$

The specific type of correlation defined in equation 33 should be maximized.

For first canonical variate pair, (U1, V1):

The coefficients $a_{11}, a_{12}, \dots, a_{1p}$ and $b_{11}, b_{12}, \dots, b_{1q}$ should be selected to maximize the canonical correlation ρ_{*1} of the first canonical variate pair, with the constraint that variances of the two canonical variates in that pair are equal to one.

$$\text{var}(U_1) = \text{var}(V_1) = 1 \quad (34)$$

This is required so that unique values for the coefficients are obtained. [80]

For the second canonical variate pair: (U2, V2)

Likewise, we want the coefficients $a_{21}, a_{22}, \dots, a_{2p}$ and $b_{21}, b_{22}, \dots, b_{2q}$ that maximize the canonical correlation ρ_{*2} of the second canonical variate pair, (U2, V2). However, in addition to the constraint that variances of the two canonical variates in this pair be equal to one, we require the additional constraints that (U1, U2), and (V1, V2) have to be uncorrelated. In addition, the combinations (U1, V2) and (U2, V1) must be uncorrelated. In summary, the overall constraints become:

$$\left. \begin{aligned} \text{var}(U_2) &= \text{var}(V_2) = 1 \\ \text{Cov}(U_1, U_2) &= \text{Cov}(V_1, V_2) = 0 \\ \text{Cov}(U_1, V_2) &= \text{Cov}(U_2, V_1) = 0 \end{aligned} \right\} \quad (35)$$

Basically, we require that all of the remaining correlations equal zero. This procedure is repeated for each pair of canonical variates. By applying this dimension reduction technique, one can find linear combinations of X's (one group of properties describing biomass) that are correlated with linear combinations of Y's (another group of properties describing biomass).

Our method in using CCA analysis to amalgamate different properties of biomass into a whole is multifaceted and involves a preconditioning step. Specifically, first, an internal linkage is made between members of each group. This is done by picking the variable with least variation coefficient in each group as the base of normalization (i.e. C% for elemental, FC for proximate, and Hem for structural). This is followed by a logarithmic transformation of ratios to set up a one-to-one correspondence between the elements of the groups (i.e. a group isomorphism).

Next, the three groups of elementals, proximate and structural are linked together performing “two” separate CCA (i.e. the minimum number of edges to ensure connectedness of 3 vertices). Overall, we form some sort of association between different types of biomass that has two particular qualities: 1. It is fully chained, and 2. It has hierarchy.

Conceptually, these steps may be viewed as depicted in Figure 27.

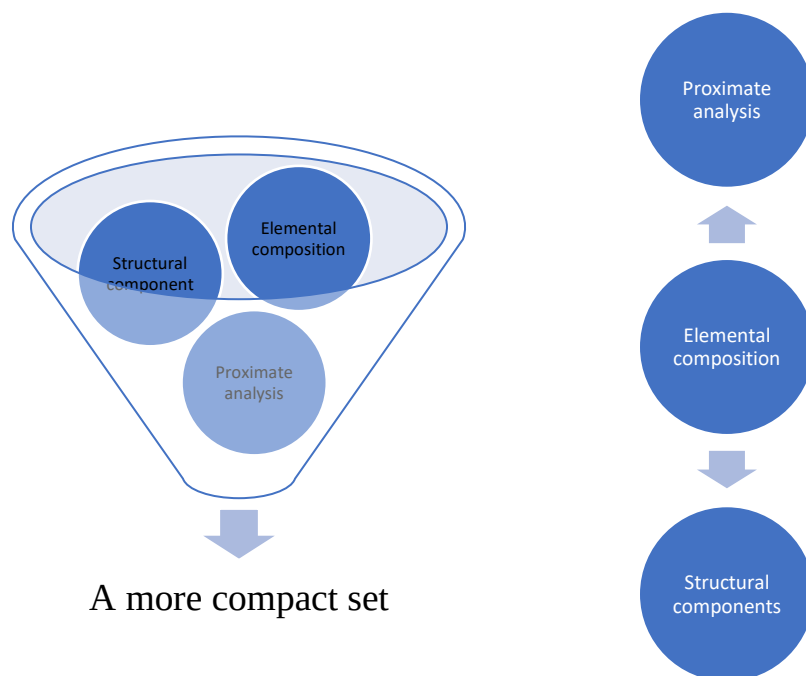


Figure 27. Biomass representative array [E1, E2, P1, S1]

Where Elemental Composition Quotient (E), Proximate Composition Quotient (P), Structural Composition Quotient (S) are indices representing their class or perspective on the feedstock. The following are the transformed composition for the elemental, proximate (dry basis) and structural components (Figures 28-29).

In our study, 38 types of biomass are considered as the sample space of the biomass system, and their properties are expressed using vectors.

	Name	Ln C/O	Ln C/H	Ln C/N	Ln C/S
1	Beech bark	0.21	2.15	4.30	6.15
2	Birch bark	0.47	2.14	4.74	6.35
3	Eucalyptus bark	0.07	2.15	5.09	6.88
4	Oak sawdust	0.13	2.14	6.22	8.52
5	Oak wood	0.17	2.12	5.13	6.23
6	Olive wood	0.09	2.21	4.25	7.40
7	Pine bark	0.30	2.21	5.19	6.64
8	Pine chips	0.27	2.16	4.66	6.37
9	Pine pruning	0.23	2.11	4.64	8.55
10	Pine sawdust	0.17	2.14	6.23	8.54
11	Spruce bark	0.29	2.16	6.28	6.28
12	Spruce wood	0.24	2.15	5.16	6.26
13	Wood	0.12	2.10	6.21	6.72
14	Wood residue	0.20	2.13	4.63	6.47
15	Bamboo whole	0.20	2.32	4.87	7.17
16	Reed canary grass	0.15	2.06	3.49	5.80
17	Sweet sorghum grass	0.13	2.10	4.82	6.31
18	Switchgrass	0.14	2.10	4.26	6.11
19	Barley straw	0.12	2.08	4.26	5.94
20	Oat straw	0.09	2.10	4.58	6.41
21	Rape straw	0.09	2.03	4.57	6.18
22	Rice straw	0.15	2.17	3.91	5.75
23	Straw	0.09	2.16	3.89	5.93
24	Wheat straw	0.12	2.09	4.26	5.67
25	Almond shells	0.17	2.09	3.92	6.91
26	Coconut shells	0.17	2.21	6.24	6.24
27	Cotton husks	0.24	1.79	3.58	8.53
28	Hazelnut shells	0.21	2.24	3.61	7.16
29	Olive husks	0.17	2.09	3.44	6.91
30	Olive pits	0.29	2.08	3.87	6.63
31	Olive residue	0.54	2.31	3.73	5.54
32	Rice husks	0.12	2.09	4.12	6.42
33	Sunflower husks	0.16	2.22	3.82	7.43
34	Walnut blows	0.40	2.10	3.67	6.21
35	Walnut hulls and blows	0.41	2.11	3.54	6.13
36	Walnut shells	0.16	2.09	3.57	6.32
37	Wood-agricultural resid	0.24	2.17	4.88	7.18
38	Wood-straw residue	0.22	2.10	4.86	5.99

Figure 28. Log transformation of elemental composition based on C

	Name	Ln(FC/VM)	Ln(FC/A)	Ln (Hem/Cell)	Ln (Hem/Lig)
1	Beech bark	-1.38	0.87	-0.32	0.39
2	Birch bark	-1.40	2.24	-0.43	0.66
3	Eucalyptus bark	-1.52	1.28	-1.23	-0.73
4	Oak sawdust	-1.86	3.68	-0.62	1.12
5	Oak wood	-1.29	3.69	-0.62	1.12
6	Olive wood	-1.53	1.68	-0.02	-0.72
7	Pine bark	-1.10	2.56	-0.72	-0.19
8	Pine chips	-1.21	1.29	-0.72	-0.19
9	Pine pruning	-1.70	1.73	-0.72	-0.19
10	Pine sawdust	-1.60	4.96	-0.55	-0.05
11	Spruce bark	-1.15	2.00	-0.46	-0.06
12	Spruce wood	-1.49	3.53	-0.62	-0.09
13	Wood	-1.68	4.28	-0.65	-0.49
14	Wood residue	-1.55	1.12	-0.72	0.00
15	Bamboo whole	-1.54	2.94	-0.50	-0.11
16	Reed canary grass	-1.43	0.69	-0.23	1.21
17	Sweet sorghum grass	-1.45	1.34	-0.72	0.00
18	Switchgrass	-1.71	1.05	-0.24	1.09
19	Barley straw	-1.41	1.25	-0.49	0.31
20	Oat straw	-1.78	0.84	-0.29	0.43
21	Rape straw	-1.47	1.33	-0.86	0.05
22	Rice straw	-1.42	-0.26	-0.47	0.79
23	Straw	-1.54	0.37	-0.29	0.40
24	Wheat straw	-1.42	0.93	-0.29	0.40
25	Almond shells	-1.24	1.87	-0.56	0.35
26	Coconut shells	-1.16	1.96	-0.37	-0.14
27	Cotton husks	-1.46	1.66	-1.58	5.14
28	Hazelnut shells	-1.28	2.65	0.12	-0.37
29	Olive husks	-1.44	2.11	-0.02	-0.72
30	Olive pits	-1.35	1.86	-0.02	-0.72
31	Olive residue	-0.97	1.27	-0.02	-0.72
32	Rice husks	-1.18	0.07	-0.33	0.25
33	Sunflower husks	-1.29	1.91	-0.70	0.59
34	Walnut blows	-1.57	1.97	-0.05	-0.53
35	Walnut hulls and blows	-1.52	1.80	-0.05	-0.53
36	Walnut shells	-0.45	2.61	-0.05	-0.53
37	Wood-agricultural resid	-1.46	1.71	-0.43	0.29
38	Wood-straw residue	-1.50	0.71	-0.29	0.43

Figure 29. Log transformation of proximate analysis based on FC, and structural component based on Hem

Here, we plug our “logarithmic variables” of Elemental and Proximate, in the set of equations (28), as X and Y, respectively.

$$U_1^{elem} = a_{11} \ln \ln \left(\frac{C}{O} \right) + a_{12} \ln \ln \left(\frac{C}{H} \right) + a_{13} \ln \ln \left(\frac{C}{N} \right) + a_{14} \ln \ln \left(\frac{C}{S} \right) \quad (36)$$

$$U_2^{elem} = a_{21} \ln \ln \left(\frac{C}{O} \right) + a_{22} \ln \ln \left(\frac{C}{H} \right) + a_{23} \ln \ln \left(\frac{C}{N} \right) + a_{24} \ln \ln \left(\frac{C}{S} \right) \quad (37)$$

...

$$V_1^{prox} = b_{11} \ln \ln \left(\frac{FC}{VM} \right) + b_{12} \ln \ln \left(\frac{FC}{A} \right) \quad (38)$$

$$V_2^{prox} = b_{21} \ln \ln \left(\frac{FC}{VM} \right) + b_{22} \ln \ln \left(\frac{FC}{A} \right) \quad (39)$$

Using basic logarithm rules, the mathematical representation of (E, P) reduces to the following:

$$U_1^{elem} = \ln \ln \left(\frac{C}{O} \right)^{a_{11}} + \ln \ln \left(\frac{C}{H} \right)^{a_{12}} + \ln \ln \left(\frac{C}{N} \right)^{a_{13}} + \ln \ln \left(\frac{C}{S} \right)^{a_{14}} =$$

$$\ln \ln \left[\left(\frac{C}{O} \right)^{a_{11}} \left(\frac{C}{H} \right)^{a_{12}} \left(\frac{C}{N} \right)^{a_{13}} \left(\frac{C}{S} \right)^{a_{14}} \right] \quad (40)$$

$$V_1^{prox} = \ln \ln \left(\frac{FC}{VM} \right)^{b_{11}} + \ln \ln \left(\frac{FC}{A} \right)^{b_{12}} = \ln \ln \left[\left(\frac{FC}{VM} \right)^{b_{11}} \left(\frac{FC}{A} \right)^{b_{12}} \right] \quad (41)$$

Where the powers of our statistical ensembles are obtained from performing canonical correlation analysis of 38 types of biomass using XLSTAT software [88].

In a similar fashion, a relationship between Elemental vs Structural components can be obtained.

The results of CCA for Elemental vs Proximate are as following (Table 15):

Where $F1 = (E_1, P_1)$

Table 15. Coefficients obtained from CCA analysis for $F1 = (E_1, P_1)$

Standardized canonical coefficients (Y1)		
	F1	
Y11		0.167
Y12		0.011
Y13		0.635
Y14		0.651
Standardized canonical coefficients (Y2)		
	F1	
Y21		-0.382
Y22		0.903

Therefore, the powers of the following relationship are obtained (Table 16), yielding the correlation depicted in Figure 30.

$$U_1^{elem} = \ln \ln \left[\left(\frac{C}{O} \right)^{a_{11}} \left(\frac{C}{H} \right)^{a_{12}} \left(\frac{C}{N} \right)^{a_{13}} \left(\frac{C}{S} \right)^{a_{14}} \right]$$

$$V_1^{prox} = \ln \ln \left[\left(\frac{FC}{VM} \right)^{b_{11}} \left(\frac{FC}{A} \right)^{b_{12}} \right]$$

Table 16. Powers of (elemental, proximate) variate

Coefficient	Value
a_{11}	0.167
a_{12}	0.011
a_{13}	0.635
a_{14}	0.651
b_{11}	-0.382
b_{12}	0.903

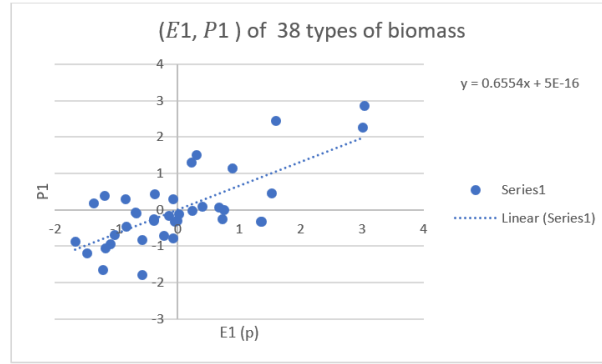


Figure 30. Elemental character vs Proximate character (E1 vs P1)

Therefore, given the elemental composition and proximate analysis (E1 vs P1) is calculated as:

$$E_1 = \frac{C^{1.45}}{O^{0.17}N^{0.63}S^{0.65}} \quad (42)$$

$$P_1 = \frac{FC^{0.52}VM^{0.38}}{A^{0.9}} \quad (43)$$

It should be noted, in case the proximate analysis of a fuel was unavailable, the parameter P_1 should be estimated based on the relation in Figure 31.

The results of CCA for Elemental vs Structural component (Table 17):

Where $F2 = (E_2, S_1)$

Table 17. Coefficients obtained from CCA analysis for $F1 = (E_1, P_1)$

Standardized canonical coefficients (Y1)		
	F2	
Y11		-0.010
Y12		-0.878
Y13		0.233
Y14		0.342
Standardized canonical coefficients (Y2)		
	F2	
Y21		-0.484
Y22		0.653

The powers of the following relationship are obtained (Table 18), yielding the correlation depicted in Figure 34.

$$U_1^{elem} = a_{11} \ln \ln \left(\frac{C}{O} \right) + a_{12} \ln \ln \left(\frac{C}{H} \right) + a_{13} \ln \ln \left(\frac{C}{N} \right) + a_{14} \ln \ln \left(\frac{C}{S} \right)$$

$$V_1 = b_{11} \ln \ln \left(\frac{Hem}{Cell} \right) + b_{12} \ln \ln \left(\frac{Hem}{Lig} \right)$$

Table 18. Powers of (elemental, structural) variate

a_{11}	-0.010
a_{12}	-0.878
a_{13}	0.233
a_{14}	0.342
b_{11}	-0.484
b_{12}	0.653

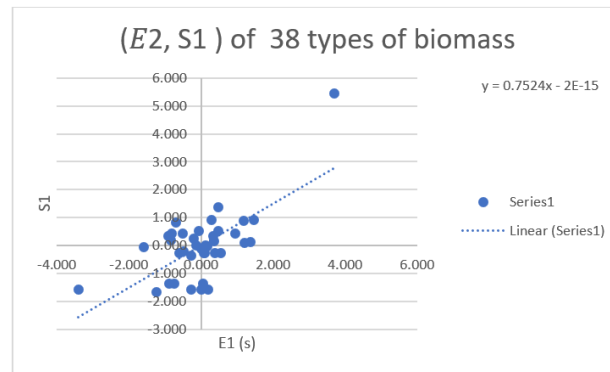


Figure 31. Elemental character vs Proximate character (E1vs P1)

Where:

$$E_2 = \ln\left(\frac{H^{0.88}}{C^{0.31}N^{0.23}S^{0.34}}\right) \quad (44)$$

$$S_1 = \ln\left(\frac{Hem^{0.17}Cell^{0.48}}{Lig^{0.65}}\right) \quad (45)$$

In this setup, the relationship between E1 and E2 is as depicted in Figure 32.

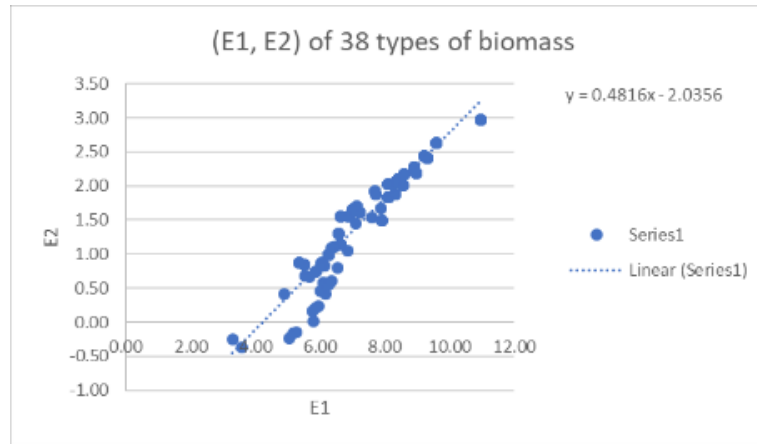


Figure 32. Elemental character (p) vs Elemental character (s) (E1vs E2)

These relationships are considered as the building blocks of our biomass characterization.

4.2 Trigonometric/ Geometric transformations

In the next step, we use trigonometric identities to develop an assortment of variables representing product gas composition.

In trigonometry, the basic relationship between the sine and the cosine is given by the Pythagorean identity.

$$\sin^2\theta + \cos^2\theta = 1 \quad (46)$$

On the other hand, sum of mole fractions of different components in product gas mixture is also one based on Daltons law.

$$\Sigma \quad y_{H_2} + y_{CH_4} + y_{CO} + y_{CO_2} = 1 \quad (47)$$

Considering the following 3 combinations:

$$\sin^2 \theta_1 = y_{H_2} + y_{CH_4} \quad \cos^2 \theta_1 = y_{CO} + y_{CO_2} \quad (48)$$

$$\sin^2 \theta_2 = y_{H_2} + y_{CO} \quad \cos^2 \theta_2 = y_{CH_4} + y_{CO_2} \quad (49)$$

$$\sin^2 \theta_3 = y_{H_2} + y_{CO_2} \quad \cos^2 \theta_3 = y_{CH_4} + y_{CO} \quad (50)$$

Knowing $\tan^2 \theta_i = \frac{\sin^2 \theta_i}{\cos^2 \theta_i}$ and using the inverse of the tangent function, we may

build the following representative vector of product gas composition for each dataset:

$$\theta_i = [\theta_1, \theta_2, \theta_3], i = 1, 2, \dots, n$$

Note that each element of this vector “single-handedly” represents the total composition of product gas. Therefore, the whole vector contains information about the “same event” from 3 different points of view. Then, by assuming each θ as a side of a triangle, we may construct the triangle representative of the product gas composition. Figure 33. illustrates progress from θ (in radian) on the unit circle to the triangle representative.

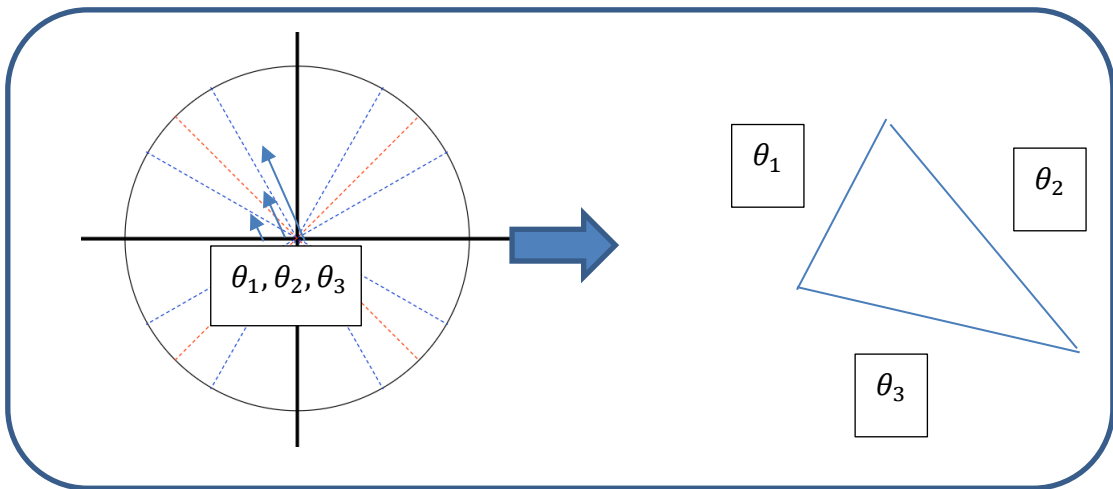


Figure 33. Illustrates progress from θ (in radian) on the unit circle to the triangle

This transformation enables us to choose from a very rich variety of trig functions in unit circle and also many well-established properties of triangles. For example, *circumcenter* is the point of concurrency of the perpendicular bisectors of a triangle; *centroid* is the point of concurrency of the medians of a triangle; *orthocenter* is the point of concurrency of the altitudes in a triangle. Although, theoretically, all of the trigonometric functions of an angle θ can be constructed geometrically (Fig. 34), we only use few of them according to our need in each analysis.

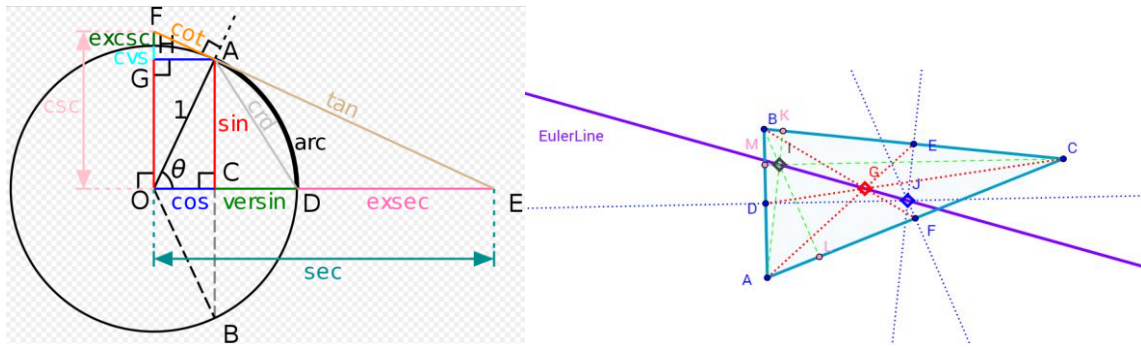


Figure 34. Graphical illustration of trigonometric functions of an angle θ and triangle centers [81]

The calculations of $\theta = [\theta_1, \theta_2, \theta_3]$ and other parameters for the first 50 datasets is given in Figures 35-36.

N° of data set										Radian			sum(theta)
	H2+CH4	CO+CO2	H2+CO	CH4+CO2	H2+CO2	CH4+CO							
1	0.38	0.62	0.64	0.36	0.41	0.59	0.60	1.79	0.71	0.66	0.93	0.70	2.29
2	0.42	0.58	0.68	0.32	0.47	0.53	0.72	2.09	0.90	0.70	0.97	0.76	2.43
3	0.44	0.56	0.70	0.30	0.50	0.50	0.80	2.33	1.02	0.73	0.99	0.79	2.51
4	0.47	0.53	0.71	0.29	0.55	0.45	0.90	2.42	1.22	0.76	1.00	0.83	2.59
5	0.48	0.52	0.73	0.27	0.59	0.41	0.91	2.70	1.41	0.76	1.02	0.87	2.66
6	0.45	0.55	0.69	0.31	0.49	0.51	0.83	2.26	0.97	0.74	0.98	0.78	2.50
7	0.44	0.56	0.70	0.30	0.48	0.52	0.80	2.32	0.91	0.73	0.99	0.76	2.48
8	0.44	0.56	0.71	0.29	0.48	0.52	0.79	2.42	0.91	0.73	1.00	0.76	2.49
9	0.43	0.57	0.70	0.30	0.48	0.52	0.77	2.34	0.94	0.72	0.99	0.77	2.48
10	0.43	0.57	0.69	0.31	0.51	0.49	0.76	2.27	1.04	0.72	0.99	0.79	2.50
11	0.41	0.59	0.77	0.23	0.49	0.51	0.70	3.31	0.98	0.70	1.07	0.78	2.54
12	0.43	0.57	0.70	0.30	0.46	0.54	0.77	2.34	0.86	0.72	0.99	0.75	2.46
13	0.44	0.56	0.70	0.30	0.48	0.52	0.78	2.35	0.92	0.72	0.99	0.76	2.48
14	0.44	0.56	0.70	0.30	0.50	0.50	0.79	2.33	1.02	0.73	0.99	0.79	2.51
15	0.44	0.56	0.70	0.30	0.51	0.49	0.78	2.28	1.04	0.72	0.99	0.80	2.51
16	0.42	0.58	0.71	0.29	0.47	0.53	0.73	2.49	0.90	0.71	1.01	0.76	2.47
17	0.43	0.57	0.72	0.28	0.48	0.52	0.76	2.53	0.94	0.72	1.01	0.77	2.49
18	0.43	0.57	0.69	0.31	0.51	0.49	0.75	2.25	1.05	0.71	0.98	0.80	2.49
19	0.42	0.58	0.69	0.31	0.53	0.47	0.71	2.21	1.12	0.70	0.98	0.81	2.49
20	0.29	0.71	0.63	0.37	0.54	0.46	0.40	1.67	1.17	0.57	0.91	0.82	2.30
21	0.32	0.68	0.65	0.35	0.50	0.50	0.47	1.88	0.99	0.60	0.94	0.78	2.32
22	0.23	0.77	0.53	0.47	0.66	0.34	0.30	1.13	1.91	0.50	0.82	0.94	2.26
23	0.35	0.65	0.76	0.24	0.42	0.58	0.53	3.15	0.73	0.63	1.06	0.71	2.39
24	0.37	0.63	0.68	0.32	0.52	0.48	0.60	2.15	1.08	0.66	0.97	0.80	2.43
25	0.32	0.68	0.67	0.33	0.51	0.49	0.46	2.02	1.03	0.60	0.96	0.79	2.35
26	0.19	0.81	0.51	0.49	0.61	0.39	0.23	1.06	1.58	0.45	0.80	0.90	2.14
27	0.38	0.62	0.59	0.41	0.63	0.37	0.62	1.42	1.71	0.67	0.87	0.92	2.46
28	0.40	0.60	0.60	0.40	0.62	0.38	0.67	1.51	1.64	0.69	0.89	0.91	2.48
29	0.40	0.60	0.60	0.40	0.61	0.39	0.67	1.50	1.54	0.69	0.89	0.89	2.46
30	0.40	0.60	0.56	0.44	0.67	0.33	0.68	1.27	2.00	0.69	0.84	0.96	2.49
31	0.36	0.64	0.61	0.39	0.61	0.39	0.57	1.55	1.56	0.65	0.89	0.90	2.44
32	0.38	0.62	0.61	0.39	0.57	0.43	0.61	1.56	1.32	0.66	0.90	0.85	2.41
33	0.36	0.64	0.60	0.40	0.62	0.38	0.57	1.51	1.65	0.65	0.89	0.91	2.44
34	0.37	0.63	0.60	0.40	0.58	0.42	0.58	1.51	1.38	0.65	0.89	0.86	2.40
35	0.30	0.70	0.62	0.38	0.52	0.48	0.44	1.61	1.09	0.58	0.90	0.81	2.29
36	0.28	0.72	0.68	0.32	0.45	0.55	0.40	2.09	0.83	0.56	0.97	0.74	2.27
37	0.27	0.73	0.64	0.36	0.52	0.48	0.37	1.75	1.08	0.55	0.92	0.80	2.27
38	0.28	0.72	0.64	0.36	0.49	0.51	0.39	1.78	0.98	0.56	0.93	0.78	2.27
39	0.30	0.70	0.66	0.34	0.53	0.47	0.42	1.90	1.11	0.58	0.94	0.81	2.33
40	0.32	0.68	0.66	0.34	0.50	0.50	0.47	1.98	1.01	0.60	0.95	0.79	2.34
41	0.27	0.73	0.65	0.35	0.51	0.49	0.37	1.84	1.04	0.55	0.94	0.80	2.28
42	0.29	0.71	0.65	0.35	0.51	0.49	0.40	1.87	1.04	0.57	0.94	0.80	2.30
43	0.34	0.66	0.65	0.35	0.58	0.42	0.51	1.83	1.37	0.62	0.93	0.86	2.42
44	0.29	0.71	0.55	0.45	0.65	0.35	0.41	1.20	1.83	0.57	0.83	0.93	2.33
45	0.21	0.79	0.45	0.55	0.72	0.28	0.27	0.80	2.55	0.48	0.73	1.01	2.22
46	0.32	0.68	0.51	0.49	0.58	0.42	0.46	1.04	1.35	0.60	0.80	0.86	2.25
47	0.23	0.77	0.37	0.63	0.70	0.30	0.30	0.60	2.30	0.50	0.66	0.99	2.15
48	0.26	0.74	0.45	0.55	0.66	0.34	0.36	0.82	1.98	0.54	0.74	0.95	2.23
49	0.27	0.73	0.56	0.44	0.55	0.45	0.36	1.27	1.21	0.54	0.85	0.83	2.22
50	0.30	0.70	0.56	0.44	0.61	0.39	0.44	1.27	1.56	0.58	0.85	0.90	2.32

Figure 35. Computation of representative vector of product gas (first 50 datasets)

s (semiperimeter)	K (area)	L (eq)	R (circumradius)	r (inradius)
1.143	0.230	0.729	0.465	0.201
1.214	0.265	0.782	0.487	0.218
1.254	0.284	0.810	0.501	0.227
1.297	0.309	0.845	0.512	0.238
1.329	0.324	0.865	0.525	0.244
1.251	0.284	0.810	0.498	0.227
1.241	0.276	0.798	0.499	0.222
1.244	0.275	0.798	0.502	0.221
1.241	0.275	0.797	0.499	0.222
1.248	0.281	0.806	0.498	0.225
1.272	0.271	0.791	0.535	0.213
1.230	0.268	0.787	0.498	0.218
1.241	0.275	0.797	0.499	0.222
1.254	0.284	0.810	0.501	0.227
1.253	0.284	0.810	0.499	0.227
1.237	0.268	0.787	0.504	0.217
1.247	0.275	0.796	0.506	0.220
1.247	0.281	0.805	0.498	0.225
1.247	0.281	0.805	0.497	0.225
1.152	0.230	0.729	0.464	0.200
1.162	0.234	0.735	0.473	0.201
1.129	0.203	0.684	0.472	0.180
1.196	0.215	0.704	0.547	0.179
1.217	0.262	0.778	0.490	0.216
1.174	0.236	0.739	0.480	0.201
1.072	0.178	0.642	0.450	0.166
1.229	0.276	0.799	0.483	0.225
1.240	0.284	0.810	0.486	0.229
1.232	0.281	0.806	0.482	0.228
1.244	0.283	0.808	0.492	0.227
1.218	0.270	0.790	0.480	0.222
1.207	0.268	0.787	0.474	0.222
1.222	0.271	0.792	0.482	0.222
1.201	0.264	0.781	0.472	0.220
1.147	0.231	0.731	0.460	0.202
1.134	0.207	0.691	0.485	0.183
1.137	0.219	0.711	0.464	0.192
1.133	0.218	0.709	0.465	0.192
1.165	0.233	0.733	0.474	0.200
1.171	0.236	0.739	0.478	0.202
1.140	0.218	0.709	0.469	0.191
1.151	0.225	0.720	0.471	0.195
1.211	0.261	0.776	0.482	0.216
1.166	0.233	0.734	0.472	0.200
1.112	0.164	0.615	0.544	0.147
1.127	0.229	0.728	0.446	0.204
1.074	0.148	0.585	0.550	0.138
1.114	0.198	0.676	0.479	0.177
1.110	0.215	0.705	0.443	0.194
1.162	0.238	0.742	0.463	0.205

Figure 36. Computation of other useful parameters of the product gas triangle

$$K_p(\text{area}) = \sqrt{s(s - \theta_1)(s - \theta_2)(s - \theta_3)} \text{ where } s \text{ is semiperimeter of triangle} \quad (51)$$

$$K_p(\text{area}) = \frac{L^2\sqrt{3}}{4} \text{ (equivalent length of an equilateral triangle with area of } K) \quad (52)$$

$$K_p = rs = \frac{\theta_1\theta_2\theta_3}{R}, \text{ where } r \text{ is inradius and } R \text{ is circumradius of the triangle} \quad (53)$$

In a similar fashion, we would like to construct the vector representative of the elemental composition, proximate analysis and structural component of biomass in a triangle form, however we need to follow a slightly different procedure, since we are consolidating 3 different groups.

Based on the correlations developed in part 1, we have Figure 37.

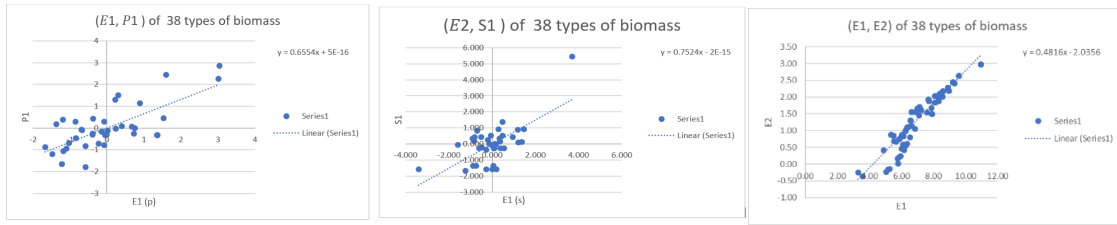


Figure 37. Correlations developed in part 1

$$E_1 = \ln\left(\frac{C^{1.45}}{O^{0.17}N^{0.63}S^{0.65}}\right), P_1 = \ln\left(\frac{FC^{0.52}VM^{0.38}}{A^{0.9}}\right),$$

$$E_2 = \ln\left(\frac{H^{0.88}}{C^{0.31}N^{0.23}S^{0.34}}\right), S_1 = \ln\left(\frac{Hem^{0.17}Cell^{0.48}}{Lig^{0.65}}\right),$$

The value of the $EPS = [E_1, E_2, S_1, P_1]$ is calculated for all the datasets. Two triangles of $\Delta E_1 E_2 S_1$ with centroid G_s and $\Delta E_1 E_2 P_1$ with centroid G_p are constructed (Fig. 38).

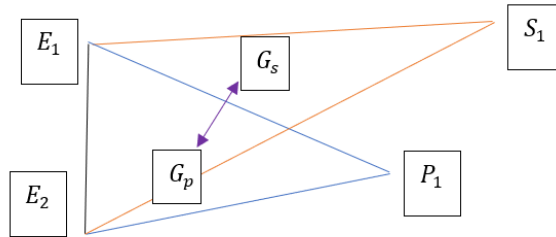


Figure 38. $\Delta E_1 E_2 S_1$ & $\Delta E_1 E_2 P_1$

We calculate the distance (d) between two centroids, and use it as a relative measure on how the structural component of biomass and its proximate analysis are

positioned with respect to its elemental composition. Therefore, G_{ps} is our first critical parameter in this regard.

The second parameter selected is the area of triangle that the base E_1E_2 makes with the middle point of $G_p - G_s$ line and denoted as (i.e. the colored area in Figure 39).

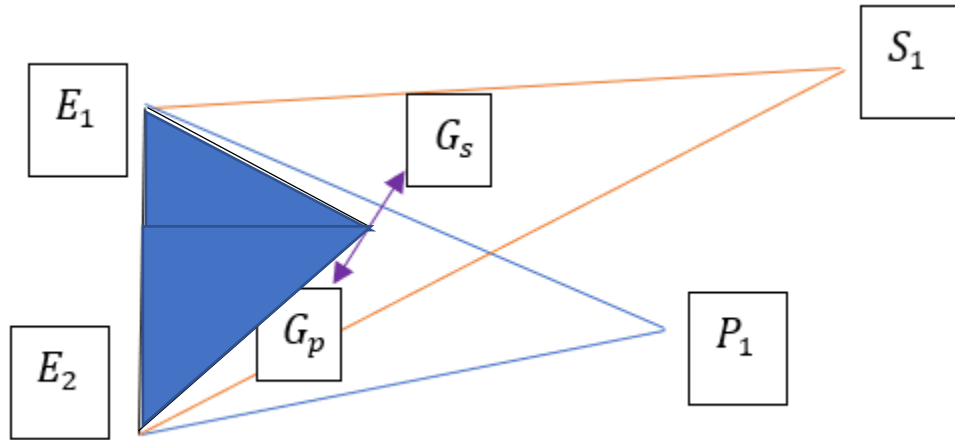


Figure 39. Illustration of the second parameter of biomass character (shaded area)

This parameter incorporates the distance between E_1E_2 in our plane which can be viewed as an enhanced packet of information regarding the elemental composition. Furthermore, the area nature of the parameter (d^2) also implies how much each data set is extended or spread overall.

Let K be the area of the triangle whose vertices are given by $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$K = \frac{1}{2} |x_1y_2 + x_2y_3 + x_3y_1 - x_2y_1 - x_3y_2 - x_1y_3| \quad (54)$$

The sample of computations is shown for the first 50 cases in Figure 40.

N° of data set	E1	E2	P1	S1	Gps	K
1	6.62	1.13	5.46	0.85	1.54	2.12
2	6.62	1.13	5.46	0.85	1.54	2.12
3	6.62	1.13	5.46	0.85	1.54	2.12
4	6.62	1.13	5.46	0.85	1.54	2.12
5	6.62	1.13	5.46	0.85	1.54	2.12
6	6.62	1.13	5.46	0.85	1.54	2.12
7	6.62	1.13	5.46	0.85	1.54	2.12
8	6.62	1.13	5.46	0.85	1.54	2.12
9	6.62	1.13	5.46	0.85	1.54	2.12
10	6.62	1.13	5.46	0.85	1.54	2.12
11	6.62	1.13	5.46	0.85	1.54	2.12
12	6.62	1.13	5.46	0.85	1.54	2.12
13	6.62	1.13	5.46	0.85	1.54	2.12
14	6.62	1.13	5.46	0.85	1.54	2.12
15	6.62	1.13	5.46	0.85	1.54	2.12
16	6.62	1.13	5.46	0.85	1.54	2.12
17	6.62	1.13	5.46	0.85	1.54	2.12
18	6.62	1.13	5.46	0.85	1.54	2.12
19	6.62	1.13	5.46	0.85	1.54	2.12
20	9.21	2.44	5.11	1.83	1.09	3.08
21	9.21	2.44	5.11	1.83	1.09	3.08
22	9.21	2.44	5.11	1.83	1.09	3.08
23	9.21	2.44	5.11	1.83	1.09	3.08
24	9.21	2.44	5.11	1.83	1.09	3.08
25	9.21	2.44	5.11	1.83	1.09	3.08
26	9.21	2.44	5.11	1.83	1.09	3.08
27	9.58	2.63	5.05	1.97	1.02	3.22
28	9.58	2.63	5.05	1.97	1.02	3.22
29	9.58	2.63	5.05	1.97	1.02	3.22
30	9.58	2.63	5.05	1.97	1.02	3.22
31	7.92	1.50	3.29	1.12	0.72	1.76
32	7.92	1.50	3.29	1.12	0.72	1.76
33	7.92	1.50	3.29	1.12	0.72	1.76
34	7.92	1.50	3.29	1.12	0.72	1.76
35	5.52	0.84	1.99	0.63	0.45	0.74
36	5.52	0.84	1.99	0.63	0.45	0.74
37	5.52	0.84	1.99	0.63	0.45	0.74
38	5.52	0.84	1.99	0.63	0.45	0.74
39	8.95	2.19	4.43	1.64	0.93	2.65
40	8.95	2.19	4.43	1.64	0.93	2.65
41	8.95	2.19	4.43	1.64	0.93	2.65
42	8.95	2.19	4.43	1.64	0.93	2.65
43	6.09	0.57	3.47	0.43	1.01	1.26
44	6.09	0.57	3.47	0.43	1.01	1.26
45	6.09	0.57	3.47	0.43	1.01	1.26
46	8.57	2.02	6.97	1.51	1.82	3.56
47	8.57	2.02	6.97	1.51	1.82	3.56
48	8.57	2.02	6.97	1.51	1.82	3.56
49	8.32	1.88	5.04	1.41	1.21	2.65
50	8.32	1.88	5.04	1.41	1.21	2.65

Figure 40. Computation of representative vector of biomass (first 50 datasets)

4.3 Construction of representative group

In the next step, a scheme with a conditional objective is implemented. The objective is to dampen high numerical variations of parameters describing events with the condition of retaining the identity of individual events. Therefore, the most important task is to hit the right level of homogeneity as a whole and independence of each case. It is necessary to establish an equal base from the starting point if we are to satisfy the homogeneity aspect. Therefore, we initiate the transformation from a circle with area equal to one ($S = 1, r = \sqrt{\frac{1}{\pi}}$) resembling a typical probability density function which is nonnegative everywhere and its integral over the entire space is equal to one. The transformative procedure for input set (i.e. Biomass properties and operating condition) is as follows:

The total area is divided into 4 sections, on the account of presence of 4 entities in the input set (i.e. biomass as a whole plus the three independent operating condition parameters, temperature (T), equivalent air ratio (ER), and steam to biomass ratio (SBR)). That said, allocation of a section to an entity is not exclusive for all, and distribution of area shares is done with following considerations:

1. The extent of involvement of a specific parameter to constitute an area is commonsurate with its significance based on the subject knowledge of the process. For example, temperature, in addition to having its own designated section, plays an auxiliary role in other sections.
2. The distribution is set up to possess the attribute of true randomness. The aim of increasing randomness intentionally, and at the earliest step of the process is to produce immunity for the model against high levels of uncertainty.

3. A lower and upper bound in calculation of each section share is considered to prevent dominance of any section. In our case, elimination of SBR section is avoided for cases where $SBR = 0$ to keep from alternating between 3 and 4 sections.
4. An internal hierarchy among sections is devised in terms of inclusiveness (Fig. 41).

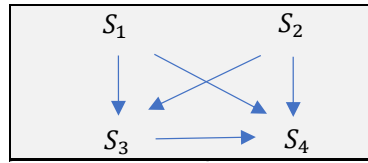


Figure 41. S_1, S_2, S_3 , and S_4 are influenced respectively by [only SBR], [T or T & ER], [Biomass, T, SBR or ALL], and [ALL]

Based on the scale of our numerical values for the input set $[d, k, T, ER, SBR]$, the following formulation and arrangement seemed to meet our conditions.

$$s_1 = \frac{3 \cdot SBR}{100} + 0.05 \quad (55)$$

$$s_2 = 0.7 * \max \left\{ \frac{T-1000}{T}, \frac{T-1000}{T} + ER \right\} \quad (56)$$

$$s_3 = \left(\frac{1+d^2}{1+d^2+\frac{10^3 k^2}{T}} \right) * (1 - (s_1 + s_2)) \quad (57)$$

$$s_4 = 1 - (s_1 + s_2 + s_3) \quad (58)$$

Transformation of a typical set of $[d, k, ER, SBR, 1000/T]$ to $[S_1, S_2, S_3, S_4]$

(Figure 42).

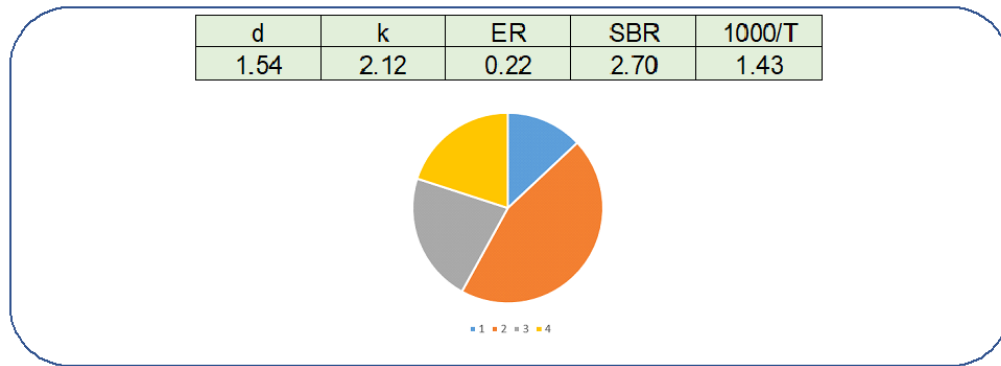


Figure 42. Illustration of the example $[s_1 = 0.13, s_2 = 0.45, s_3 = 0.22, s_4 = 0.20]$

The representation illustrated in Figure 30 has several advantages as discussed, however in following our strategy to enforce an effective system of checks and balances preventing the unfairly favoring certain subdivisions, it is decided to make use of the wellknown structure of a bicameral legislature system in which each state is represented both based on “population or size” and also “regardless of the size”. Here, we retain the property of size (i.e. area of circular sector) and for the second basis we assign “exactly one full quadrant” to each section regardless of their size (Fig. 43). Geometrically, the similarity of each partition and perpendicularity of lines separating the four areas are desirable features.

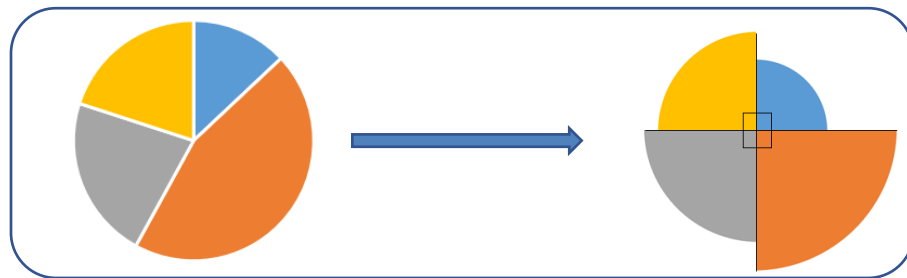
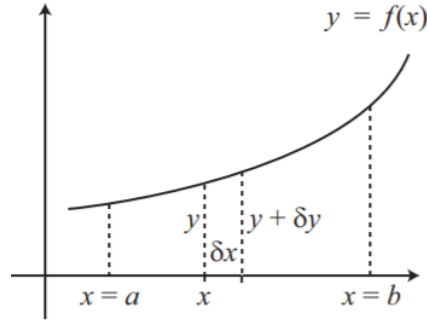


Figure 43. Assign “exactly one full quadrant”

Then, the revolutionary solid volume of quadrants is calculated to turn the 2D information to 3D to attain full geometric representation in terms of “number of dimensions”. In general we have:



$$V = \sum_{x=a}^{x=b} \partial V = \int_a^b \pi y^2 dx \quad (59)$$

Sum of the volumes for each quadrant with respect to the x and y axis yields three quarters of an sphere with the same radius.

Therefore:

for $i = 1, 2, 3, 4$ $r_i = 20 * \sqrt{\frac{S_i}{\pi}}$ (multiplied by 20 only for scaling numerical values), and

$$v_i = \pi r_i^3,$$

Another advantage of the transformation from slices with different angles to full quadrants is that quadrants can be divided into a straight segment (i.e. an isosceles right triangle) and a curvilinear segment (i.e. a cord with a fixed arc length) only by addition of one line (this is viewed as efficient binary classification).

Since the role of curvilinear segment is naturally have been addressed via the volumetric variable (v_i), we turn our attention to the straight segment which exhibited a strong relationship with its counterpart from output set (i.e. the product gas composition).

The product of ratio of altitudes ($l_i = \frac{\sqrt{2}}{2} r_i$) in right triangles (Fig. 44) showed 72%

correlation with the same variable built from output set. This strong correlation implies the efficacy of the steps taken up to this point in building highly informative characteristics.



Figure 44. Altitudes and binary areas of the kite

Additional averaging operations were done.

$$\text{Sum of the equivalent radius } R_j = \sum_1^4 r_i, \quad (60)$$

$$\text{Sum of the areas made out of binary altitudes } S_j = \frac{l_1 l_2}{2} + \frac{l_2 l_3}{2} + \frac{l_3 l_4}{2} + \frac{l_4 l_1}{2} \quad (61)$$

$$\text{A weighted sum of volumes: } V_j = \min \left(273 + \frac{v_1}{16} + \frac{v_2}{28} + \frac{v_3}{2} + \frac{v_4}{44}, 940 \right) \quad (62)$$

It can be seen that the selected weights are actually the molecular weights of the product gas composition. This type of assignment was implemented based on the assumption that we would gain more through this great leap of increasing similarity between two groups of input and output than what we may lose due to lack of an inherently one to one relation between these volumes and gas components. An analogy to this situation is where a 1v1 defending tactic is used in soccer alongside the overall strategy of defending as a group.

$$\text{Ratio of diagonals of the kite: } L_j = \frac{l_1}{l_4} \quad (63)$$

$$\text{Composite variable: } N_j = \frac{1}{2} \left(\frac{L_j * V_j}{T_j S_j * \ln(R_{j_max} - R_j)} + 12 \right) * \frac{2.3 * R}{F}, \quad (64)$$

where T_j is the temperature in °C, R =the universal gas constant, and F is the Faraday constant.

The rationale for involvement of Faraday constant is the following. To make the mathematical object created in the previous process “electrically charged” with the aim of enabling or possibly enhancing its frequency response when tested with respect to the variable of “computational time”.

In the last step of “representative” preparation we aim to recuperate its overall “uniformity” which may have been weakened due to multiplexity of steps taken after the equity circle ($S=1$).

For this purpose, the structure of Fibonacci sequence is utilized which has a prominent feature that $\frac{f_{n+1}}{f_n} = \varphi$, where φ is the golden ratio ($\frac{1+\sqrt{5}}{2}$).

The integers in Fibonacci numbers are constructed as following, starting with $f_1 \& f_2 \& f_3$ as seed values, and then building upon them by the convention that n th Fibonacci number is the sum of the previous two Fibonacci numbers:

$$f_1 = N_j + e^{-N_j}, f_2 = f_1 + e^{-N_j}, f_3 = e^{-N_j} * f_2, f_4 = f_3 + f_2, f_5 = f_4 + f_3, f_6 = f_5 + f_4, f_7 = f_6 + f_5, f_8 = f_7 + f_6, f_9 = f_8 + f_7, f_{10} = f_9 + f_8, f_{11} = f_{10} + f_9 \quad (65)$$

The golden ratio (approximately 1.6) starts to emerge after f_6 based on our computations and gets closer to actual value of ($\frac{1+\sqrt{5}}{2}$) moving forward. Therefore, we selected the set of $F_j = [f_6, f_7, f_8, f_9, f_{10}, f_{11}]$ as our final representatives that will go through the functional analyses. (It should be noted for convenience the subscripts 1 to 5

is used instead of 6 to 11 from now on) Figure 45 summarizes the major steps taken to obtain what we now perceive as a cohesive representation.

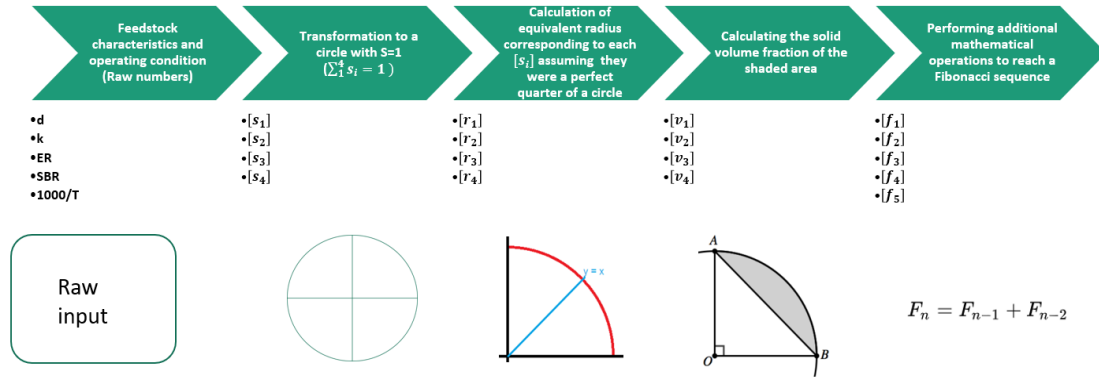


Figure 45. The progress of representative preparation

The following tables show few examples of final sets built from original data

(Figure 46):

f1	f2	f3	f4	f5
16	26	42	69	111
15	24	40	64	104
15	24	39	62	101
15	24	39	62	101
15	25	40	64	104
16	26	42	69	111
16	26	42	69	111
16	26	43	69	112
16	27	43	70	113
17	27	43	70	114
20	32	53	85	137
17	27	44	71	114
15	25	40	65	106

: Fibonacci sequence
obtained from Given
information of the
problem (Feed&
Operating condition)

Figure 46. Examples of Fibonacci sequences constructed from composite variable N_j .
(Page 1 of 2)

f1	f2	f3	f4	f5
18	29	47	76	123
18	30	48	77	125
21	35	56	90	146
23	37	59	96	155
27	43	70	114	184
21	34	55	89	144
22	36	58	93	151
24	39	63	102	164
23	37	60	96	156
22	36	58	93	151
30	49	80	129	208
23	37	59	96	155
22	35	57	91	148

: Fibonacci sequence
obtained (using
similar procedure)
from synthesis gas
composition (H₂,
CO, CO₂ and CH₄)

34	55	89	145	234
33	54	87	141	229
36	58	94	153	247
37	60	98	158	256
42	68	110	178	288
37	60	97	158	255
38	62	100	162	262
40	65	105	171	276
39	63	103	166	269
39	62	101	163	265
50	82	132	214	346
39	64	103	166	269
37	60	97	157	254

+

Figure 46. Examples of Fibonacci sequences constructed from composite variable N_j.
(Page 2 of 2)

The following plots illustrate how the distance between center of A and B differs among the 200 cases under study (First plot: a typical example, Second plot: an abnormal instance). The distance between two centers is of critical importance (Fig. 47).

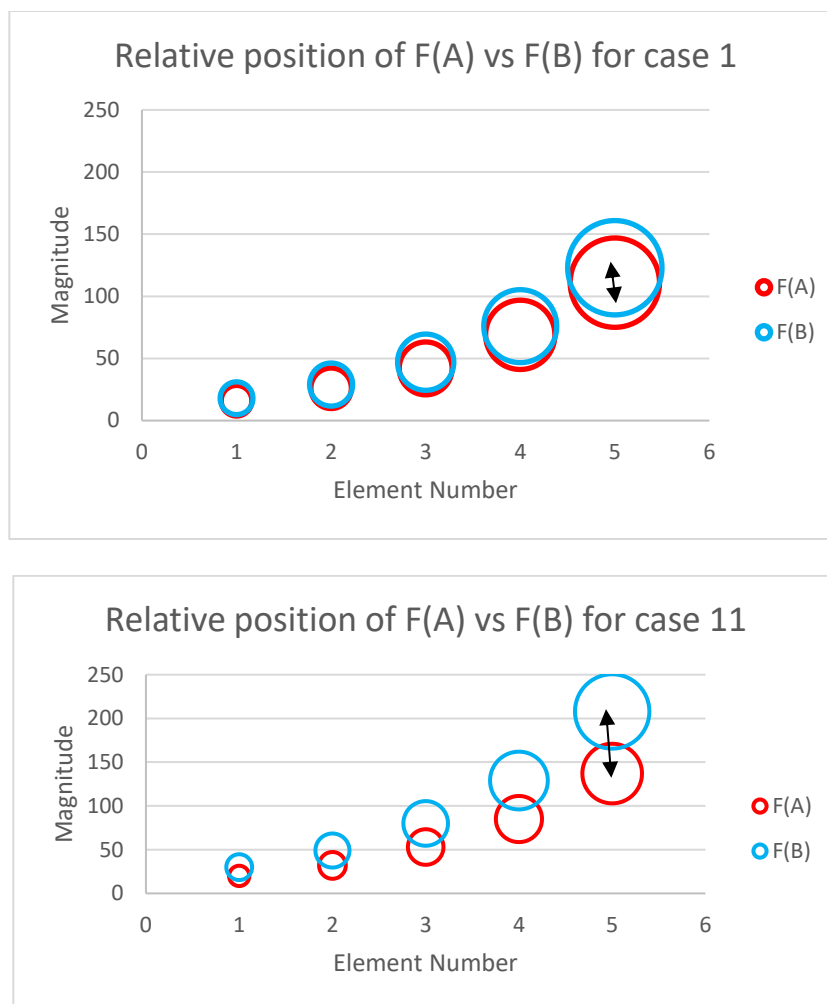


Figure 47. The centric differences of two associated Fibonacci sequences, up: typical, bottom: abnormal

At this juncture, we have successfully converted our original problem into harmonious integer sequences, representing the event (i.e. feedstock, operating condition, and product gas composition). It should be noted that carbon conversion efficiency (CCE%) and product gas yield (Nm³/kg) are the output variables that should be predicted by the model, and not included as part of the event embedded in the sequences. That will keep us two degrees of freedom away from “totality” of the process. The overall transformations practically meet the specifications of a formal mathematic “Group”,

defined in Group theory: “In mathematics, a group is an algebraic structure consisting of a set of elements equipped with an operation that combines any two elements to form a third element and that satisfies four conditions called the group axioms, namely closure, associativity, identity and invertibility. One of the most familiar examples of a group is the set of integers together with the addition operation, but the abstract formalization of the group axioms, detached as it is from the concrete nature of any particular group and its operation, applies much more widely. It allows entities with highly diverse mathematical origins in abstract algebra and beyond to be handled in a flexible way while retaining their essential structural aspects. The ubiquity of groups in numerous areas within and outside mathematics makes them a central organizing principle of contemporary mathematics.”[82]

The next step is to design a mathematical set up to serve us as a computational labrotary. Then by performing specific analyses which will be elaborated shortly, we will predict the most probable outcome of biomass gasification in fluidized bed reactor for a given input set.

The central idea of our statistical approach is “indirectness”, therefore basing the inference about the identity of an inquiry on some type of performance along an arbitrary path. The inclusiveness of the designed path and stages upon which the performance is to be monitored has direct effect on the quality and amount of information we gain. In our study, a closed, multilateral path is laid out starting from Fibbonacci sequences, followed by series of matrix operation, and ultimately reaching a fully symmetric destination (similar to the invariant point in geometric projections) that reveals accurate and deep understanding about the original problem.

4.4 Design of neural based computational lab

Here, we first give a brief and general introduction on neural networks, and then explain what role they play in our modeling.

Neural networks are conceptually quite simple computing systems, inspired by structure of biological neurons in human brain. Neural networks use computational methods to “learn” information from data without relying on a predetermined equation as a model. Deep learning is especially suited for image recognition, which is important for solving problems such as facial recognition, motion detection. “Such systems progressively learn through examples without task specific programming. Various architectures of ANNs are being used of which Multilayer Perceptron (MLP) is the most popular one. MLP consists of an input layer, a hidden layer and an output layer of neurons. While the neurons in the input layer are typically linear, the neurons in the hidden layer have nonlinear activation functions. The neurons in the output layer may be linear or nonlinear. The input layer consists of input neurons with associated weight. The hidden layer receives the signals from the input layer. There’s also a threshold parameter known as bias which is associated with each neuron in the hidden and output layers” [83]. “An elementary neuron with R inputs is shown below. Each input is weighted with an appropriate w . The sum of the weighted inputs and the bias forms the input to the transfer function f . (Fig. 48)” [84]

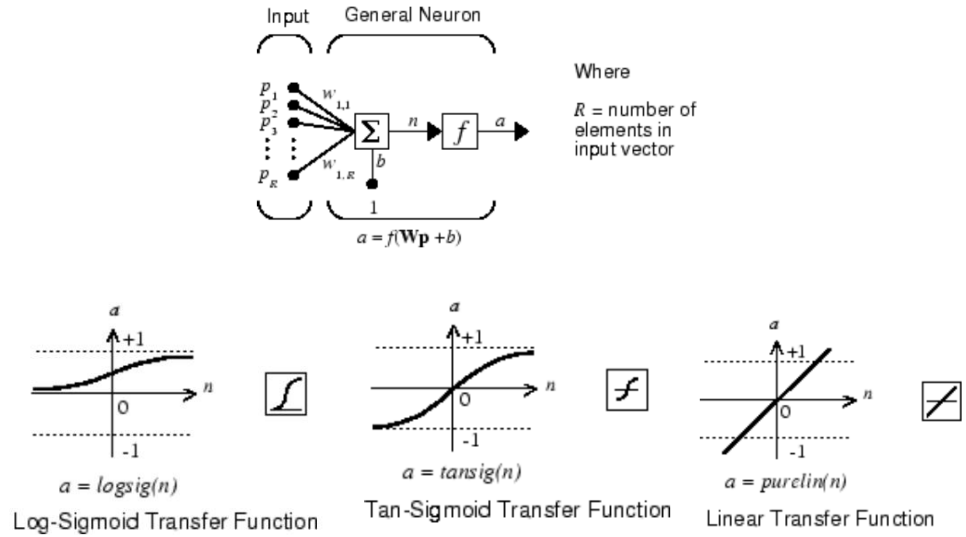
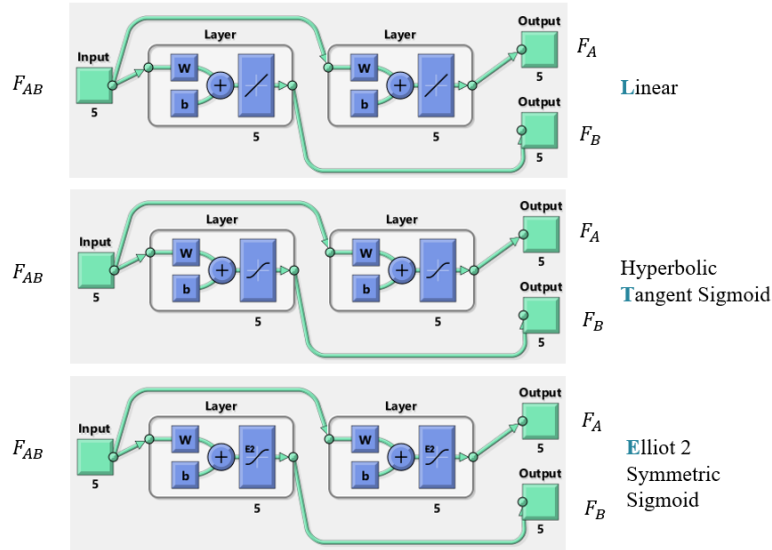


Figure 48. The structure of a general neuron as foundational unit of neural networks [84]

Development of three stable neural network unit featuring vanished gradients to serve as a computational laboratory:

Three neural network unit with three different transfer functions (i.e. linear, hyperbolic tangent sigmoid, and elliot symmetric sigmoid) were trained by the sets of Fibonacci sequences with the configurations illustrated in Figure 49. (The ABT is chosen to exemplify the NN details).



- All NN's are composed of an input layer that receives F_{AB} , 2 hidden layers each with 5 neuron designated to predict F_A and F_B separately.
- **ABL**, **ABT**, **ABE** are 3 NN's trained based on **L**inear, **T**angential and **E**lliptical transfer function, respectively.

- There are **60** elements in ABT.
- **50** elements in terms of weight (w) (i.e. one matrix of $[]_{5 \times 5}$ for each layer).
- And, **10** elements in terms of bias (b) (i.e. one vector of $[]_{5 \times 1}$ for each layer).
- For example, the following numerical configuration, is resulted upon training the "ABT" neural network using Levenberg-Marquardt backpropagation algorithm with aim of minimizing sum of squared errors in predicting F_A and F_B .

-1.42	2.51	1.77	1.28	1.35	-0.81
-0.61	0.48	-3.08	0.94	-1.98	3.09
-2.01	-2.19	-2.27	-0.24	0.93	2.89
-2.20	-1.59	-1.98	0.28	-1.89	2.72
3.13	-1.04	1.06	-1.44	-0.93	1.55

1.43	-0.15	0.12	3.54	0.57	-4.69
2.42	-0.55	2.75	0.28	1.06	-3.94
-2.98	-0.38	2.10	-1.06	0.62	0.85
2.30	-1.04	0.77	2.35	-1.56	-0.44
2.04	0.06	-0.90	2.79	-1.47	0.67

The first layer in the figure connected to output F_B

The second layer in the figure connected to output F_A

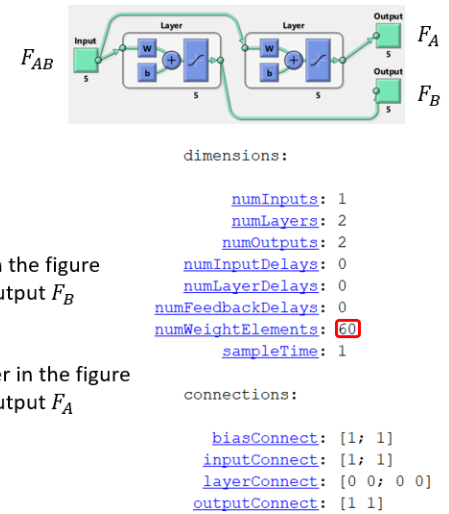


Figure 49. The detailed configuration of ABT (i.e. the neural network set up using hyperbolic tangent as transfer function)

Following the training, the neural network units shows properties of a bounded function with respect to the inputs (i.e. F_{AB}), meaning for all the input they return identical vectors of :

$$F_A = [1, 1, 1, 1, 1]; F_B = [1, 1, 1, 1, 1];$$

This seemingly short circuited configuration provides critical information on the hollistic state of data distributions. Specifically, the weights and biases calculated upon the training indirectly capture the distribution of input sets with respect to their corresponding output set. This can be viewed as an unconventional method of averaging distributions.

We will show how this particular quality can form a structure preserving map, and yield a commutative diagram across the three different neural network units.

Following our strategy to cover broad ranges of possibilities with respect to variation potentials (vertical, horizontal, and cross variations) The following scheme was arranged:

- Three different fractions of F_A (i.e. 0.4, 0.7 and 1) are considered.
- The input to ABL is in form of $[0.4 * F_A, 0.7 * F_A, F_A]$ while the input to ABT and ABE are in form of $[\frac{1}{0.4 * F_A}, \frac{1}{0.7 * F_A}, \frac{1}{F_A}]$.

Defining the quantitative measures reflecting relative deviations from:

$$F_A = [1, 1, 1, 1, 1]; F_B = [1, 1, 1, 1, 1];$$

is done with a combinatory approach using “trace” of matrices as scores. For each run of $[0.4, 0.7, 1]$ vs $[L, T, E]$ (i.e. 9 pairs) two groups of calculation is done:

1. Changes in F_B with respect to F_A , and the F_{AB}
2. Changes in F_A with respect to F_B , and F_{AB} .

It is important to notice that “changes in F_A and F_B ” exactly mean their deviations from identity vectors $F_A = [1, 1, 1, 1, 1]; F_B = [1, 1, 1, 1, 1]$.

Another important issue is that the neural network units are trained with the input set of F_{AB} where $F_{AB} = F_A + F_B$ however when we feed the

input to the neural network to monitor its changes, our input set is actually $F_{AB} = F_A$ because we do not have information about F_B . That said, the NN's behavior is unaffected and they will “act” as they are receiving some sort of F_{AB} that is constructed as $F_{AB} = F_A + F_B$. This subtle issue enables us to observe the effects of a specific F_A (i.e. an inquiry) on F_B and quantify its relation with respect to all or nearly all elements or aspects of the process.

As for the quantification method and scoring system the following matrix operation is established. All matrices have the dimension of $[]_{5 \times 5}$.

- $[B * A^T], [dB * A^T], [dB * A_{in}^T]$, (reflecting the changes of B to A) (67)
- $[A * B^T], [dA * B^T], [A_{in} * dA^T]$, (reflecting the changes of A to B) (68)

When the trace of the three matrices are summed up in each group, they yield a Figure 50 scores such as below.

		L	T	E
B to A	0.4	1.05	3.94	3.30
	0.7	1.67	3.64	3.00
	1	2.14	3.52	2.90
A to B	0.4	1.12	-1.28	3.43
	0.7	1.71	-1.12	2.94
	1	2.10	-1.04	2.70

Figure 50. Scores calculated from traces of matrices defined by Equations 67&68

Conceptually this figure summarizes the changes along 4 axioms: path, scale, AtB and BtA. Each score in table 6 is the sum of traces for three matrices ($\sum_3 []_{5 \times 5} \text{ trace}$) in aforementioned groups, it can be seen that the information in table 6 is obtained from traces of 54 matrices $[]_{5 \times 5}$ (i.e. 3 scale* 3 path* 6 measure = 54 matrices that overall consist of 1350 numerical value, each probably capturing an infinitesimal relation

between A&B) corresponding to a single inquiry. An important feature of this set up is that forms a commutative diagram between linear, hyperbolic tangent, and elliot modules as illustrated in the next Page (Fig. 51).

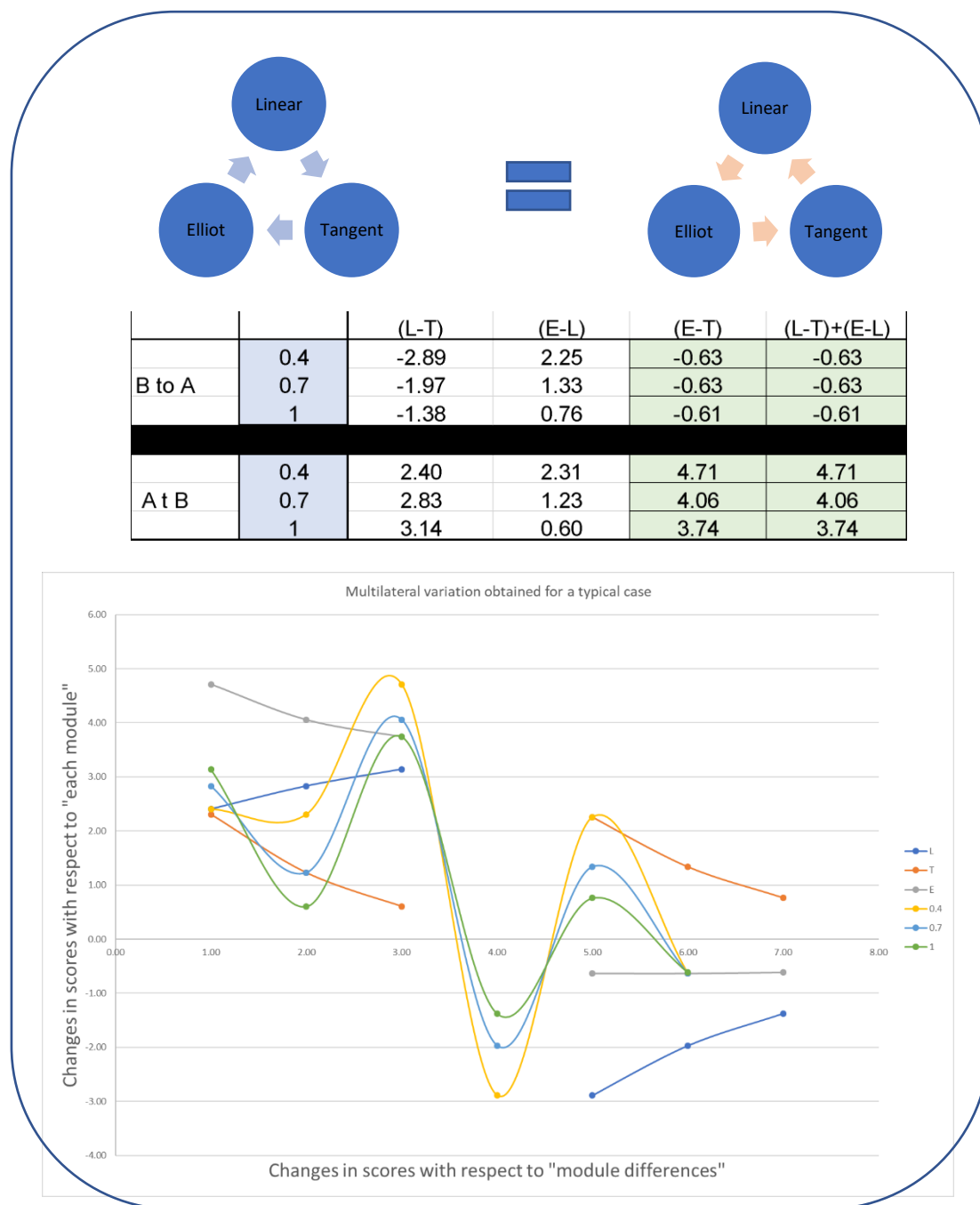


Figure 51. Elliot modules as illustrated

An important and inherent aspect of self-ruling functional analyses is to select a point where achievement of entanglement quantification can be fairly speculated. The other important consideration in these type of analyses is to decide the “form” and “dimension” (i.e single scalar, vector, involution, etc.) of the mathematical object ready for final integration. Here, since we are dealing with a convoluted object, it does not seem economical to perform rigorous structural analysis of each case, therefore, we base our judgement on the “general principle” of equilibrium for stationary objects. To be stationary all of the forces on the object must be in equilibrium, the resultant force and moment vectors must both be zero. From structural point of view, the number of equilibrium conditions required to solve different force systems can be summarized as following:

Type of force system	two dimensional	Three-dimensional
General	3	6
Coplanar	3	3
Concurrent	2	3
Parallel	2	3
Coplanar, parallel	2	2
Coplanar, concurrent	2	2
Collinear	1	1

Figure 52. Number of equilibrium conditions required to solve different force systems
(Civil engineering reference manual 41-7)

Given our aim to solve a general system, 6 equilibrium condition is needed. By considering all the 4 axioms, the summation shown in the last column in Figure 53 is chosen as the 6 equilibrium conditions.

		L	T	E	Sum
B to A	0.4	1.05	3.94	3.30	8.29
	0.7	1.67	3.64	3.00	8.31
	1	2.14	3.52	2.90	8.56
A to B	0.4	1.12	-1.28	3.43	3.26
	0.7	1.71	-1.12	2.94	3.54
	1	2.10	-1.04	2.70	3.75

Figure 53. Calculation of sum of rows to get 6 static condition

It can be seen that equilibrium conditions for changes of B with respect to A (i.e. the first 3 values) are close to the universal gas constant ($R=8.314$). This could be partially attributed to involvement of R in construction of the composite variable:

$$N_j = \frac{1}{2} \left(\frac{L_j * V_j}{T_j S_j * \ln \ln (R_{j_max} - R_j)} + 12 \right) * \frac{2.3 * R}{F},$$

where T_j was the temperature in °C, R =the universal gas constant, and F was the Faraday constant, however we are neither to claim that is necessarily the case nor think its correctness or incorrectness significantly reinforce or invalidate our course of actions. A similar argument can be applied with regard to closeness of equilibrium measures for changes of A with respect to B (i.e. the second 3 values) to the mathematical constant π .

That said, π & R can be used to quantify the measure of dispersion around “two” fixed values to normalize the “bimodal” distribution at hand. (In some sense, it is πR -reviewed!).

By normalization of the first 3 values with respect to $R=8.314$ and the second 3 scores with respect to $\pi = 3.14$, we get the numerical values with the following scale:

	1	2
1	[0.9974;0.9994;1.0290]	[1.0383;1.1259;1.1947]
2	[1.0047;0.9943;1.0162]	[1.0212;1.0961;1.1725]
3	[1.0143;1.0044;1.0274]	[1.0481;1.1511;1.2651]
4	[1.0175;1.0100;1.0354]	[1.0682;1.1863;1.3153]
5	[1.0277;1.0373;1.0773]	[1.1419;1.3093;1.4805]
6	[1.0101;1.0216;1.0608]	[1.0757;1.1913;1.2882]
7	[1.0213;1.0411;1.0887]	[1.0719;1.1846;1.2786]
8	[1.0253;1.0469;1.0928]	[1.1151;1.2655;1.4035]
9	[1.0271;1.0555;1.1081]	[1.0824;1.2032;1.3057]
10	[1.0212;1.0431;1.0875]	[1.0768;1.2099;1.3256]

Figure 54. Static vectors denoted as S_{AB} & S_{BA}

Following this transformation, dispersions around 1, now is supposed to reflect the differential information among different cases. In other words, since “1” is directly associated with 2 fundamental constants π & R , and itself reflects total equity, we may establish the connection between differentials and a fundamental equilibrium state (Equity=1). At this stage, although the non-equilibrium factor in our process has been characterized statically, there remains a fundamental gap: lack of a variable with nature of Rate & Resistance.

Basically, thermodynamics considerations do not establish the rates of chemical or physical processes because resistance is not a thermodynamic variable.

Any proposition to resolve this crucial issue must effectively combine the two seemingly paradoxical concepts of static/equilibrium and dynamics.

Here, we propose a novel technique of analysis based on “abstract physical operation on matrices” with respect to “time”. Let us exemplify the procedure by first examining 4 matrices formed by average multiplication of two static vectors for cases 1-4 as shown below:

$$S_{ci} = \frac{1}{2} (S_{ABi} * S_{BAi}^T + S_{BAi} * S_{ABi}^T), \quad (69)$$

$$S_{c1} = \begin{bmatrix} 1.035 & 1.080 & 1.130 \\ 1.080 & 1.125 & 1.176 \\ 1.130 & 1.176 & 1.229 \end{bmatrix},$$

$$S_{c2} = \begin{bmatrix} 1.025 & 1.058 & 1.107 \\ 1.058 & 1.089 & 1.139 \\ 1.107 & 1.139 & 1.191 \end{bmatrix}$$

$$S_{c3} = \begin{bmatrix} 1.063 & 1.110 & 1.180 \\ 1.110 & 1.156 & 1.226 \\ 1.180 & 1.226 & 1.299 \end{bmatrix},$$

$$S_{c4} = \begin{bmatrix} 1.086 & 1.143 & 1.222 \\ 1.143 & 1.198 & 1.278 \\ 1.222 & 1.17 & 1.361 \end{bmatrix}$$

The arrows in the following matrix indicate the direction of increase in the numerical values. This uniform pattern exists in all the 200 cases, and its emergence may be attributed to the overall harmonic tone of our analyses.

$$S_{ci} = \begin{bmatrix} S_{ci}^{11} & S_{ci}^{12} & S_{ci}^{13} \\ S_{ci}^{21} & S_{ci}^{22} & S_{ci}^{23} \\ S_{ci}^{31} & S_{ci}^{32} & S_{ci}^{33} \end{bmatrix}$$

4.5 Institution of a static/dynamic variable & integration

Now, in order to obtain a dynamical property value from a static situation, there is a need for an operation consisting some sort of change without tearing or breaking the static structure of the object. A stretching process to deform the object with condition of sorting the values from lowest to highest value seems to have this particular quality.

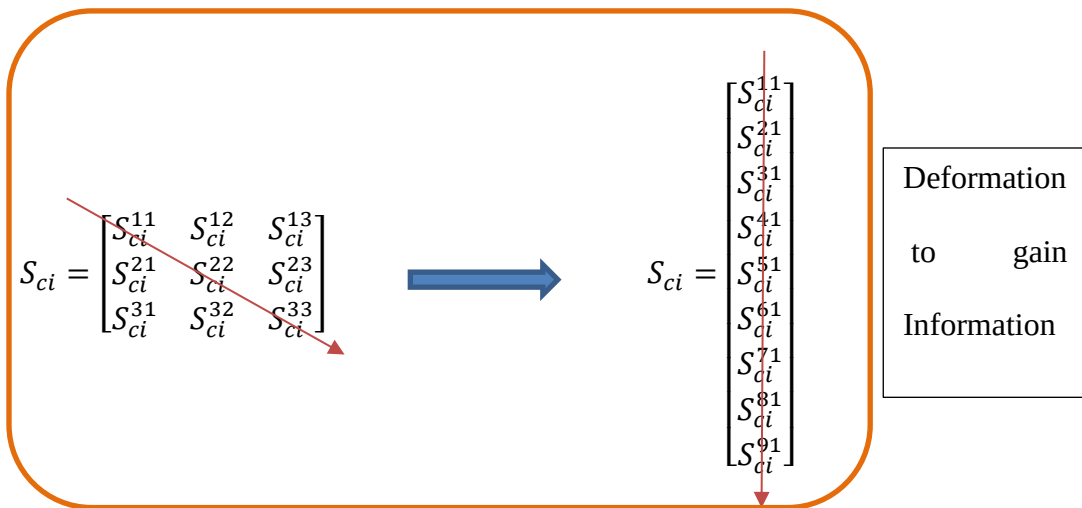


Figure 55. Deformation to gain Information

Therefore, the operation of reshaping the average product of static vectors $[]_{3 \times 3}$ to $[]_{9 \times 1}$, and arranging them from lowest to highest (as if those scores were time series or any other naturally increasing quantity such as entropy) is deemed to represent the act of “change in order”, and the duration of time from starting this specific activity till its completion is recorded. This specific computational time varied across different cases. Given the similarity of environmental condition in this test such as the device power, those differences are hypothesized to be related to stiffness of computation for different cases. This hypothesis of “relation to stiffness” is proposed based on (or inspired by) two notions:

1. The analogy between our defined event and the concepts of elasticity and strain for a physical object. (when a material is deformed under a tectonic force)
2. The theory of computational complexity that states:

“In computer science, the computational complexity, or simply complexity of an algorithm is the amount of resources required for running it. The computational complexity of a problem is the minimum of the complexities of all possible algorithms for this problem (including the unknown algorithms).

As the amount of needed resources varies with the input, the complexity is generally expressed as a function $n \rightarrow f(n)$, where n is the size of the input, and $f(n)$ is either the worst-case complexity, that is the maximum of the amount of resources that are needed for all input of size n , or the average-case complexity, that is average of the amount of resources over all input of size n . *When the nature of the resources is not explicitly given, this is usually the time needed for running the algorithm, and one talks of time complexity.* However, this depends on the computer that is used, and the time is

generally expressed as the number of needed elementary operations, which are supposed to take a constant time on a given computer, and to change by a constant factor when one changes of computer”.[85]

Specifically, we utilized an application in the MATLAB environment which allows user to record the duration time of a specific operation. The operation was done for all 200 cases and the time for each case recorded using “timeit” functionality, all the numerical values are given in appendix C.

```
For n=1:200 f = @()
tscollection(reshape(sumrows_LTE{n,3},9,1));tsc=tscollection(reshape
(sumrows_LTE{n,3},9,1));St_En(n,1)=tsc.TimeInfo.Start;
St_En(n,2)=tsc.TimeInfo.End; t1 = timeit(f); St_En(n,3)=t1; End
```

Figure 56. Function of “timeit”

An important feature that distinguishes this part of analysis from the rest, is its computational irreversibility.

“As was first argued by Rolf Landauer of IBM, in order for a computational process to be physically reversible, it must also be logically reversible. Landauer's principle is the rigorously valid observation that the oblivious erasure of n bits of known information must always incur a cost of $nkT \ln(2)$ in thermodynamic entropy. A discrete, deterministic computational process is said to be logically reversible if the transition function that maps old computational states to new ones is a one-to-one function; i.e. the output logical states uniquely determine the input logical states of the computational operation.”[86]

It is important to notice that “inverse functions” can be calculated for trigonometric, geometric transformations, the neural network formulations, etc. However, given a particular “time complexity”, the static matrix cannot be back calculated, since the relationship between them is not a bijection (Fig. 57). There is more than one configuration of that matrix with the same computational complexity.

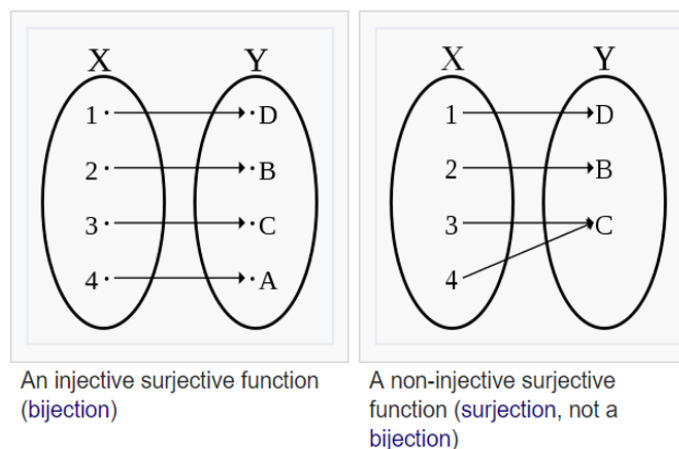


Figure 57. Surjective vs non-surjective function

Nevertheless, attainment of this dynamical/static property value proved to be extremely useful in resulting monotonic trends for important process variables such as carbon conversion efficiency and the yield of product gas (figures 14-17). The specific computational time (sct) marks the finish line of our functional analysis. In the last step, we need to perform an integration with regard to the variables we consider the most critical, and informative regarding the path traveled by our inquiry.

For the choice of integration, given the presence of two focal points in our normalization which was selected due to bimodality of the distribution, a complete “elliptic” integral of the first kind was recognized as the most compatible option. The

integration was done through a low pass filter. Specifically, the Ellipap module in MATLAB was employed. This module returns the zeros, poles, and gain of an order “n” elliptic analog lowpass filter prototype, with Rp dB of ripple in the passband, and a stopband Rs dB down from the peak value in the passband (Fig. 38) The zeros and poles are returned in length n column vectors z and p and the gain in scalar k. If n is odd, z is length n - The transfer function in factored zero-pole form returns the poles and zeros, and gain of:

$$H(S) = \frac{Z(S)}{P(S)} = k \frac{(s-z_1)(s-z_2)\dots(s-z_N)}{(s-p_1)(s-p_2)\dots(s-p_M)} \quad (70)$$

This module uses ellipke to calculate the “complete elliptic integral” of the first kind and ellipj to calculate Jacobi elliptic. It computes the Jacobi elliptic functions using the method of the arithmetic-geometric mean starting with a *triplet* of numbers. The ellipap module was fed the following three variables which were literally obtained from a *trip* along the *L*, *E*, *T* modules:

1. The specific computational time (sct)
2. The summation of 5 Fibonacci number for an inquiry (A) (i.e. starting point)
3. The absolute temperature (T)

$$a_0 = 1, b_0 = \sqrt{1-m}, c_0 = \sqrt{m}$$

$$a_i = \frac{1}{2}(a_{i-1} + b_{i-1}) \quad b_i = (a_{i-1}b_{i-1})^{\frac{1}{2}} \quad c_i = \frac{1}{2}(a_{i-1}b_{i-1})$$

$$\sin(2\phi_{n-1} - \phi_n) = \frac{c_n}{a_n} \sin(\phi_n)$$

$$\operatorname{dn}(u) = \sqrt{1-m \cdot \operatorname{sn}(u)^2} \quad \operatorname{sn}(u) = \sin \phi_0 \quad \operatorname{cn}(u) = \cos \phi_0$$

Where the Jacobi elliptic functions are

$$u = \int_0^\phi \frac{d\theta}{\sqrt{1-m \cdot \sin^2 \theta}}$$

$$\operatorname{sn}(u) = \sin \phi \quad \operatorname{cn}(u) = \cos \phi \quad \operatorname{dn}(u) = \sqrt{1-m \cdot \sin^2 \phi}$$

$$[K(m)] = \int_0^1 [(1-t^2)(1-mt^2)]^{\frac{-1}{2}} dt$$

Figure 58. The Ellipap module formulation scheme

4.6 Preliminary results

The typical results for poles, zeros and gain are shown below.

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	14x1 comp...	16x1 comp...	16x1 comp...	18x1 comp...	20x1 comp...	18x1 comp...	14x1 comp...	16x1 comp...	14x1 comp...	16x1 comp...	22x1 comp...	16x1 comp...	14x1 comp...
2	15x1 comp...	16x1 comp...	17x1 comp...	18x1 comp...	20x1 comp...	18x1 comp...	15x1 comp...	16x1 comp...	15x1 comp...	16x1 comp...	23x1 comp...	16x1 comp...	15x1 comp...
3	1.4482e-47	3.9811e-53	1.9248e-52	3.9811e-58	1.2589e-60	1.2589e-55	3.1201e-52	1.2589e-55	3.0026e-52	1.2589e-55	3.6470e-53	1.2589e-55	3.4475e-52

Figure 59. Typical results for poles, zeros and gain.

The value of gains displayed in the third row are denoted with Λ . $\ln \left(\frac{\Lambda_S}{\Lambda_G} \right)$ where Λ_G is Geomean of all cases, and Λ_S the value computed for a specific case is a dimensionless number that we call the local minima of algorithm.

The difference between size of pole and zero vectors are either 1 or 0 which we call Squaricity Index (Fig. 60&61).

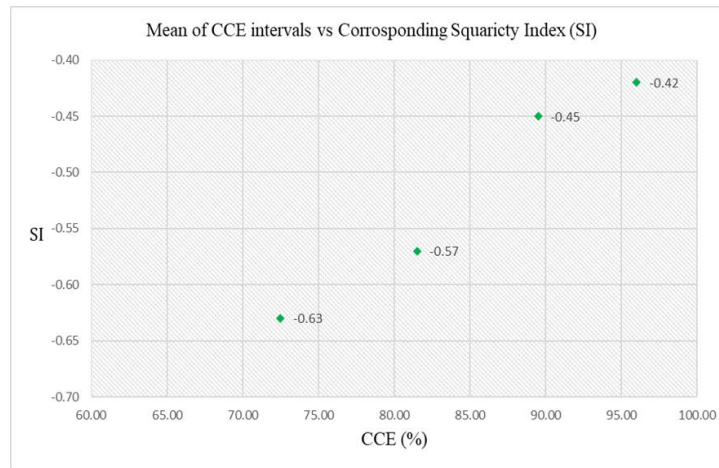


Figure 60. Carbon Conversion Efficiency vs Squaricity Index

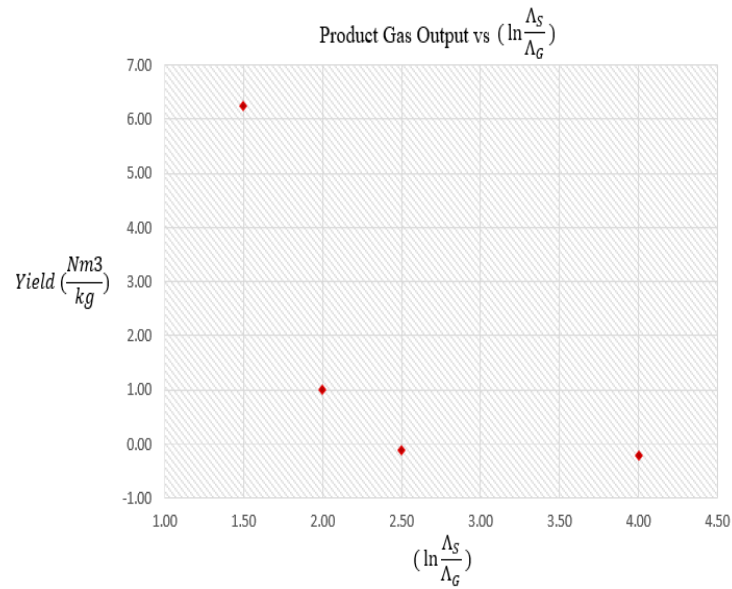
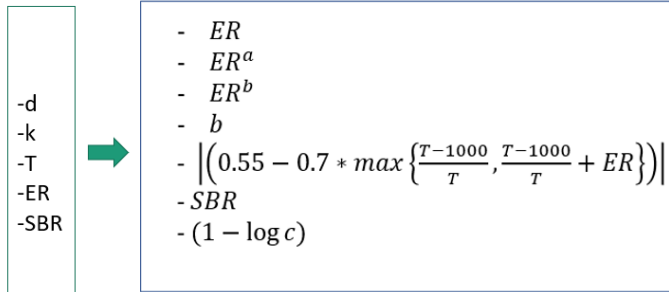


Figure 61. Syngas yield vs local minima of algorithm

For prediction of the product gas composition a neural network module was built utilizing the constructed parameters:



$$\text{Where } a = \frac{\sqrt{\left(\frac{6 * SBR + 10}{\Pi} \right)}}{c}$$

$$c = \sqrt{\left(\frac{140 * \max \left\{ \frac{T-1000}{T}, \frac{T-1000}{T} + ER \right\}}{\Pi} \right)}$$

$$b = \frac{\Pi * \left(1 + d^2 + \frac{10^3 k^2}{T} \right) * \left(1 - 0.7 * \max \left\{ \frac{T-1000}{T}, \frac{T-1000}{T} + ER \right\} - \frac{3 * SBR + 5}{\Pi} \right)}{c * 200 * (1 + d^2)}$$

$$y_{H_2} + y_{CH_4} + y_{CO} + y_{CO_2} = 1 ,$$

$$\cos^2 \theta_1 = y_{CO} + y_{CO_2} ,$$

$$\cos^2 \theta_2 = y_{CH_4} + y_{CO_2} ,$$

$$\cos^2 \theta_3 = y_{CH_4} + y_{CO} ,$$

$$\sin^2 \theta_1 = y_{H_2} + y_{CH_4} ,$$

$$\sin^2 \theta_2 = y_{H_2} + y_{CO} ,$$

$$\sin^2 \theta_3 = y_{H_2} + y_{CO_2} ,$$

$$y_{H_2} = 1 - \frac{\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3}{2} \quad (71)$$

$$y_{CH_4} = 1 - \frac{\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3}{2} - \sin^2 \theta_1 \quad (72)$$

$$y_{CO} = 1 - \frac{\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3}{2} - \sin^2 \theta_2 \quad (73)$$

$$y_{CO_2} = 1 - \frac{\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3}{2} - \sin^2 \theta_3 \quad (74)$$

Where the relationship between $[d, k]$, $[E_1, E_2, S_1, P_1]$, and $[C, H, O, N, S, Cell, Hem, Lig, Vm, Fc, A]$ is as follows:

$$d = \frac{P_1}{3} - \frac{S_1}{3} \quad (75)$$

$$k = \frac{1}{2} \left| \frac{P_1 + S_1}{6} * \frac{2E_1 - E_2}{3} \right| \quad (76)$$

$$E_1 = \ln\left(\frac{C^{1.45}}{O^{0.17}N^{0.63}S^{0.65}}\right), P_1 = \ln\left(\frac{FC^{0.52}VM^{0.38}}{A^{0.9}}\right), E_2 = \ln\left(\frac{H^{0.88}}{C^{0.31}N^{0.23}S^{0.34}}\right), S_1 = \ln\left(\frac{Hem^{0.17}Cell^{0.48}}{Lig^{0.65}}\right) \quad (77)$$

And $[T, ER, SBR]$ is the gasification operating condition.

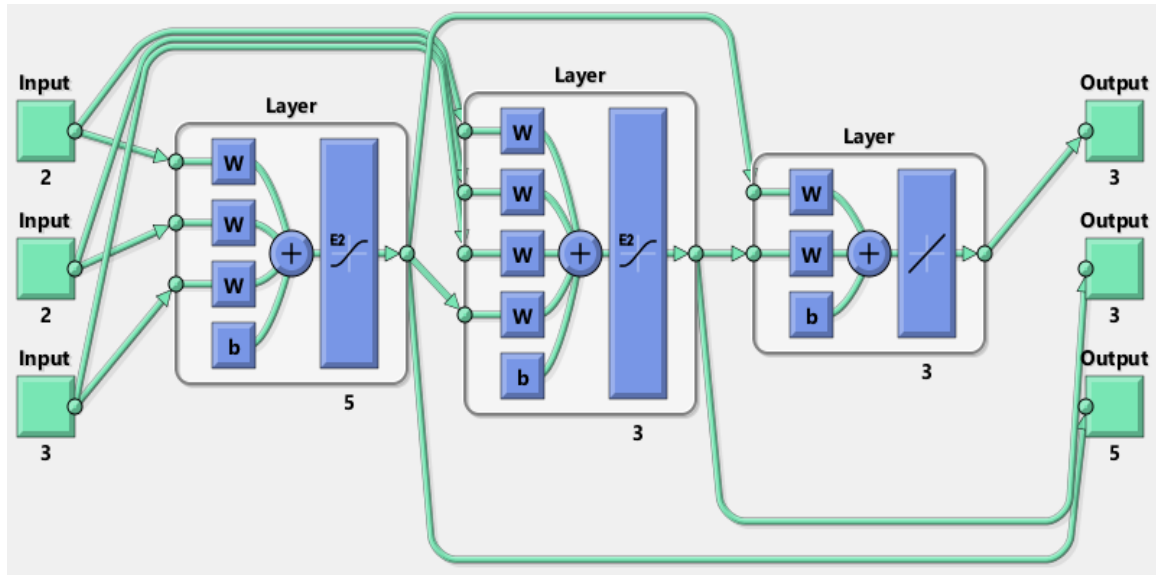
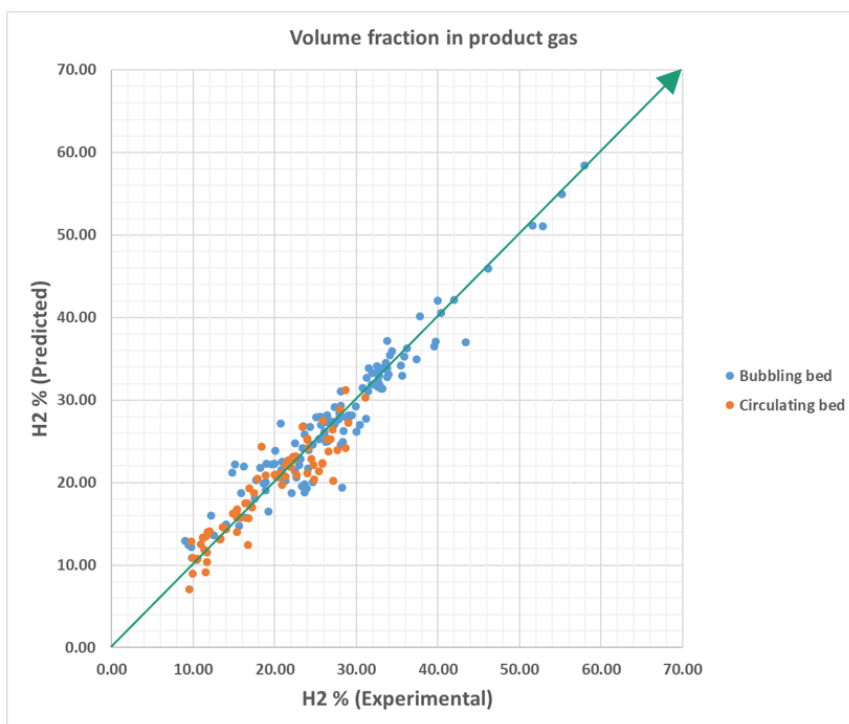
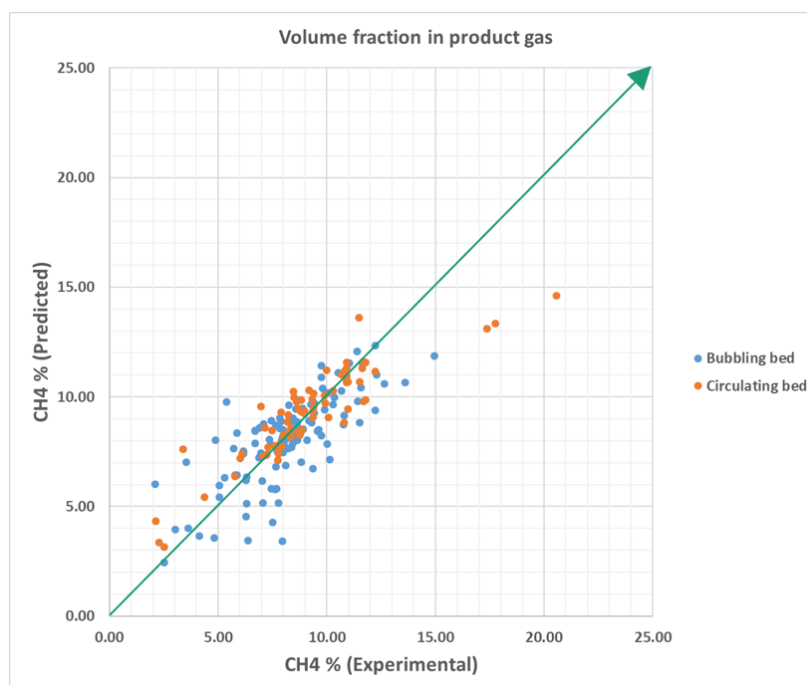


Figure 62. The module developed for prediction of product gas composition

The plots of prediction vs experimental (Overall Error=%8.6) for gas composition is presented in Figures 63-66:

Figure 63. H₂ prediction vs experimental valuesFigure 64. CH₄ prediction vs experimental values

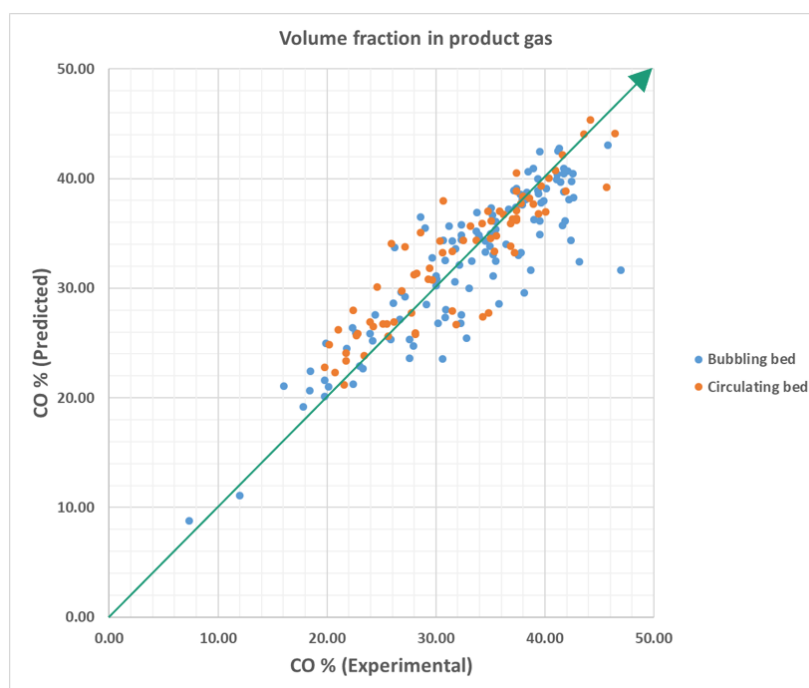


Figure 65. CO prediction vs experimental values

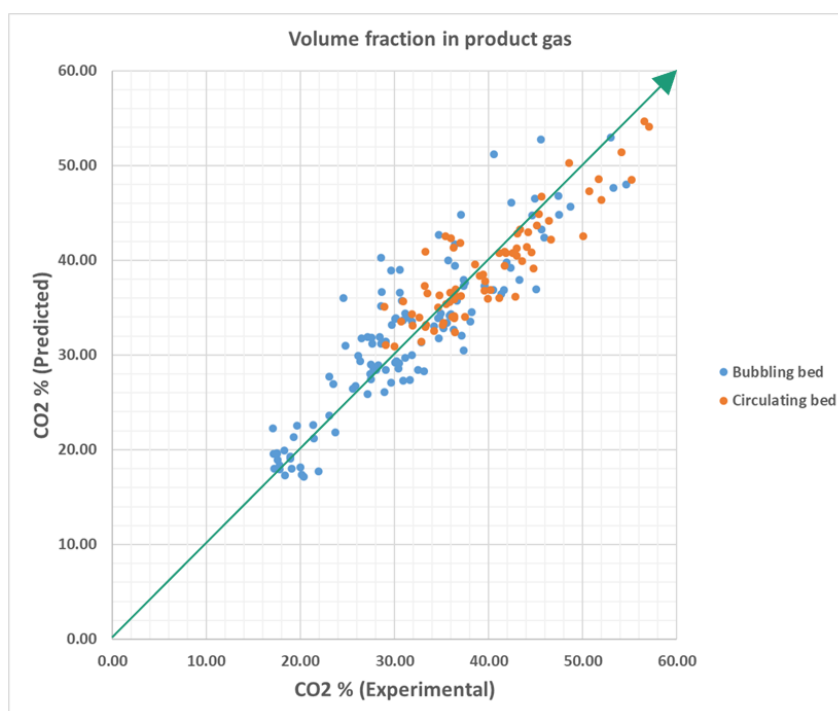


Figure 66. CO2 prediction vs experimental values

CHAPTER 5

RESULTS, DISCUSSIONS

What is presented in this chapter is essentially juxtaposition of two slightly different performance evaluation analyses, one borrowed from literature evaluating three approaches of modeling biomass gasification in fluidized bed and one done in this study in form of a sensitivity analysis on our model. The two analyses have the important common feature of evaluating models against completely new unseen datasets, a metric that is very dear to our hearts.

We consider the 114 setpoints of gas yield, 96 setpoints of carbon conversion collected from more than 15 different experiments as part of this “completely new unseen datasets” because they had not been used at any stage of our modeling effort although we had used the composition data pertinent to those gasification setpoints. It should be mentioned some modellers may have a more strict definition of completely new unseen datasets.

As for composition data, the three models evaluated by Moshtaghi [17] were tested against 63 setpoints with exception that 6 out 63 and 15 out 63 setpoints had been used in developing Model II and Model III respectively. Our model was tested against 70 new datasets.

5.1 Performance of different modeling approaches

Three approaches of modeling biomass gasification in fluidized bed “MODEL I, which restricts equilibrium to a specific temperature (QET method); MODEL II, which uses empirical correlations for carbon, CH₄, C₂H₂, C₂H₄, C₂H₆ and NH₃ conversion;

and MODEL III, which includes kinetic and hydrodynamic equationsd” was studied and compared by Moshtaghi [17].

“MODEL I, MODEL II and MODEL III (Figure 70.) are first replicated in ASPEN PLUS and verified by comparison with the model results in the original presented studies. Experimental data from different BFB and CFB gasifiers have been collected from the literature and used to evaluate the model performance, with the aim to test whether the models are also valid for experimental conditions other than those for which they were originally validated.” [17]

“The cases with input parameters in different ranges than those of the original validated experiments have been chosen for the evaluation step.” [17]

“Basically each model has been validated with exactly the same data used by the authors that developed that model. However due to further use of the input data presented in this table for evaluation of models, it is important to report all of the experimental data used for validation of the original models. Further, the experimental data that have been used for the evaluation of all models are listed in Figure 68-69. Each “setpoint” is a combination of one set of ER value, S/B ratio, temperature, pressure and load.” [17]

“To form a common basis for comparison of the different models, the same set of experimental data has been used for the evaluation of all models.” [17]

“The experimental setups chosen for evaluation of the models are listed in Figure 69. Moreover, set points **1 to 6 of CFB-EXP II** and **1 to 15 of BFB-EXP III** from Figure 68 are also used for model evaluation. These points are highlighted with italic outlined numbers in Figure 68.” [17]